

TACME: Task-Aware Cross-Modal Experts for Disjoint-Corpus Multimodal Behavioral Understanding

Elena Ryumina Alexandr Axyonov Dmitry Ryumin

HSE University, 101000 Moscow, Russia

St. Petersburg Federal Research Center of the Russian Academy of Sciences (SPC RAS),
199178 St. Petersburg, Russia

evryumina@hse.ru; aaxyonov@hse.ru; daryumin@hse.ru

Abstract

Multimodal behavioral analysis is typically studied using separate task-specific corpora. Practical applications, however, would benefit from unified models that can support several behavioral predictions despite fragmented supervision. We study this problem as disjoint-corpus multimodal multi-task learning across emotion recognition on CMU-MOSEI, apparent Big Five personality assessment on ChaLearn FIV2, and ambivalence recognition on BAH. We propose Task-Aware Cross-Modal Experts (TACME), a multimodal fusion model with a shared bank of directed cross-modal Transformer experts and task-specific expert gates. Each expert models one ordered modality interaction, while task-specific gates select different expert mixtures for different behavioral tasks. We further adapt feature- and prediction-level flow-matching regularization to this disjoint-corpus behavioral setting and propose a Progress-Aware Gradient Balancing (PAGB) multi-task learning strategy for heterogeneous task optimization. Across three independent corpora and six task-specific metrics, TACME with PAGB and feature- and prediction-level flow-matching regularization achieves the best average rank of 1.00 among the compared multimodal fusion models. Although single-task TACME is stronger on the source-corpus metrics, the multi-task TACME model improves zero-shot unweighted average recall (UAR) on the unseen MELD test set from 28.9% to 35.9%, demonstrating improved generalization to unseen data.

1 Introduction

Behavioral analysis is usually studied using separate corpora for emotion recognition (Wang et al., 2025a), personality assessment (Sandhan et al., 2025), and other psychological characteristics inferred from human behavior (Ghate et al., 2025). Practical scenarios, however, often require a single model that can jointly assess several aspects of

human behavior rather than multiple task-specific pipelines (Samiul et al., 2025). Such scenarios arise in health monitoring, personalized education, adaptive human–computer interaction, and other applications where decisions may depend on multiple behavioral and psychological characteristics.

The main obstacle is fragmented supervision. Corpora for behavioral tasks are collected independently and differ in labels, clip duration, modality quality, and evaluation metrics (Poria et al., 2019; Ma et al., 2024). An example from one corpus has labels for only one task, but a unified model should still reuse behavioral structure across tasks. Existing multimodal fusion models (He et al., 2025; Zhou et al., 2025a; Zhuang et al., 2025; He et al., 2026) provide strong baselines, but they typically learn a single fusion pattern for one target task or one corpus setting. Training across multiple corpora may improve generalization beyond the individual training corpora by exposing the model to more diverse behavioral patterns (Amiriparian et al., 2024). At the same time, joint optimization of heterogeneous tasks and corpora can introduce gradient conflicts and negative transfer (Borsani et al., 2025; Ryumina et al., 2026), reducing performance on the individual source corpora. The key challenge is therefore to share task-relevant behavioral information across corpora while limiting corpus-specific specialization and negative transfer.

To address this problem, we propose Task-Aware Cross-Modal Experts (TACME), a multimodal fusion model for this disjoint-corpus setting. Our goal is to learn a shared representation of behavioral information across independently annotated corpora while retaining task-specific cross-modal interactions. The model uses four streams: verbal content, Visual Large Language Model (VLLM)-generated behavioral descriptions, visual behavior, and acoustic/prosodic behavior. Two streams are language-based: spoken transcripts and VLLM-generated descriptions that convert visible behavior

into a textual semantic interface. Each stream is converted into a fixed two-token statistical summary and projected to a shared hidden space. A bank of directed Transformer experts then models ordered modality interactions, such as acoustic-to-visual or transcript-to-acoustic attention. The expert bank is shared across tasks, but each task learns a separate gate over the experts. This design lets the model share cross-modal interaction modules while allowing emotion, personality, and ambivalence recognition to rely on different modality relations.

We also study two mechanisms for heterogeneous Multi-Task Learning (MTL). Feature- and prediction-level Flow Matching (FM) regularize fused representations and task outputs (Jha et al., 2026). Progress-Aware Gradient Balancing (PAGB) updates task weights from validation progress, validation degradation, and smoothed gradient norms. These mechanisms address two difficulties of disjoint-corpus learning: heterogeneous output spaces and unequal task learning dynamics.

We evaluate TACME on CMU Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI), ChaLearn First Impressions v2 (FIv2), and Behavioural Ambivalence/Hesitancy (BAH), comparing fusion models, aggregation strategies, FM variants, MTL strategies, modality ablations, and task-specific systems. We further compare single-task and multi-task TACME in terms of source-corpus performance, corpus specialization, and zero-shot generalization. The results indicate that multi-task TACME trades some source-corpus performance for lower corpus specialization and improved generalization to unseen data. The main architectural contribution is the task-aware directed expert design; statistical aggregation, FM, and PAGB are evaluated as complementary components. Appendix A further clarifies their roles.

- We study joint multimodal MTL of emotion, personality, and ambivalence recognition from independently annotated corpora with heterogeneous labels, clip duration, modality quality, and evaluation metrics.
- We propose TACME, which shares directed cross-modal Transformer experts and learns task-specific expert weights, and compare it with recent multimodal fusion models under the same disjoint-corpus protocol.
- We evaluate two-token statistical aggregation and

feature- and prediction-level FM as complementary mechanisms for stabilizing heterogeneous multimodal representations.

- We introduce PAGB for validation-aware task balancing and compare it with learned weighting and established MTL strategies.

2 Related Work

Our work is related to multimodal fusion, MTL strategies, and task-specific systems.

2.1 Multimodal Fusion Models

Recent multimodal models improve the alignment and fusion of heterogeneous signals. UniGraph2 learns a shared embedding space for multimodal graphs (He et al., 2025). DPDF-LQ uses dual-path dynamic fusion with a learnable query for multimodal sentiment analysis (Zhou et al., 2025a). HME addresses missing-modality sentiment analysis through hyper-modality enhancement and uncertainty-aware fusion (Zhuang et al., 2025). ComP introduces cross-modal prompting for incomplete multimodal emotion recognition (He et al., 2026). These models are strong fusion baselines, but they are mainly designed for a single target task or corpus setting, not for joint learning from independent corpora with different label spaces.

2.2 Multi-Task Learning Strategies

MTL strategies address task imbalance by modifying losses or gradients. PCGrad (Yu et al., 2020) and CAGrad (Liu et al., 2021) reduce gradient conflict. FairGrad formulates MTL as fair resource allocation (Ban and Ji, 2024). NTKMTL studies imbalance through task convergence behavior (Qin et al., 2025), while IMGrad introduces imbalance-sensitive gradient updates (Zhou et al., 2025b). These methods operate at the optimization level; they do not specify which cross-modal relations should be shared across tasks.

2.3 Task-Specific Systems

Task-specific systems remain important reference points. For emotion recognition, recent multimodal systems combine text, audio, and visual cues through cross-attention, pooling, triple fusion, or multimodal fusion (Markitantov et al., 2026; Chen et al., 2026; Arumugam et al., 2025). For apparent personality assessment, recent methods model Big Five traits with multimodal fusion, graph relational

Transformers, multi-scale audio-visual representations, and adaptive channel selection (Ning et al., 2026; Wang et al., 2025b; Zhang et al., 2025). For ambivalence recognition, recent systems emphasize multimodal conflict, divergence-based fusion, text-guided fusion, and ensemble regularization fusion (Bekhouché et al., 2026; Souza et al., 2026; Pereira et al., 2026). These works are optimized for our research corpora and are therefore reported separately in our single-task comparison.

2.4 Position of This Work

Unlike prior fusion models, we do not assume that one fusion pattern is optimal for every task. Unlike standard MTL strategies, we do not treat task imbalance only as a loss- or gradient-level problem. TACME shares directed cross-modal experts across tasks, while task-specific gates select the modality interactions that are useful for emotion, personality, and ambivalence recognition. Additional details on the compared fusion models and MTL strategies are provided in Appendix B; task-specific systems are summarized in Appendix C.

3 Problem Formulation

Let $\mathcal{T} = \{\text{emo}, \text{per}, \text{amb}\}$ be the set of behavioral tasks, where emo, per, and amb denote emotion, apparent personality, and ambivalence recognition, respectively. For each task $t \in \mathcal{T}$, we are given a task-specific corpus

$$\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}, \quad (1)$$

where $\mathcal{D}_t$ denotes the training corpus for task t , $N_t = |\mathcal{D}_t|$ is the number of examples in that corpus, x_i^t is the i -th input example, and y_i^t is its task-specific label or target. The corpora are disjoint: an example from one corpus is not assumed to have labels for the other tasks. The target tasks are predefined, but co-annotated samples are not required. During multi-task training, a label-availability mask activates only losses with available ground-truth labels; unavailable task losses do not contribute to the update. The target spaces are heterogeneous:

$$y_i^{\text{emo}} \in \mathcal{Y}_{\text{emo}}, \quad y_i^{\text{per}} \in \mathcal{Y}_{\text{per}}, \quad y_i^{\text{amb}} \in \mathcal{Y}_{\text{amb}}, \quad (2)$$

where $\mathcal{Y}_{\text{emo}}$ is the seven-dimensional soft-label target space for emotion recognition, constructed from the original six-dimensional CMU-MOSEI emotion-intensity annotations. $\mathcal{Y}_{\text{per}} = [0, 1]^5$ contains the five normalized Big Five trait scores for

Flv2, and $\mathcal{Y}_{\text{amb}} = \{0, 1\}$ is the binary ambivalence label space for BAH.

Each input example is represented through four modality streams:

$$\mathcal{M} = \{\text{T}, \text{D}, \text{V}, \text{A}\}, \quad (3)$$

where T denotes spoken transcripts, D denotes VLLM-generated behavioral descriptions of the video, V denotes visual signals and A denotes acoustic signals. We write

$$x_i^t = \{x_{i,m}^t\}_{m \in \mathcal{M}}, \quad (4)$$

where $x_{i,m}^t$ is the stream of modality m for example i from task t . The two text-based streams play different roles: T captures what the speaker says, whereas D converts visible behavior into a textual semantic interface.

Let $f_t(\cdot; \theta)$ be the prediction function for task t , where θ contains both shared parameters and task-specific parameters. At optimization step or epoch q , the multi-task loss is

$$\mathcal{J}(\theta) = \sum_{t \in \mathcal{T}} \lambda_t^{(q)} \frac{1}{N_t} \sum_{i=1}^{N_t} \mathcal{L}_t(f_t(x_i^t; \theta), y_i^t). \quad (5)$$

Here, $\mathcal{J}(\theta)$ is the total MTL loss, N_t is the number of examples in corpus $\mathcal{D}_t$, $\lambda_t^{(q)}$ controls the contribution of task t at training step or epoch q , and $\mathcal{L}_t$ is the task-specific loss function. The task weights $\lambda_t^{(q)}$ can be equal, learned, or dynamically updated. We use cross-entropy losses for emo and amb, and mean absolute error for per.

4 Method

4.1 Overview

Figure 1 gives the overall structure. The pipeline extracts modality-specific feature matrices, summarizes each stream with mean and standard deviation (STD) tokens, and projects the summaries to a shared hidden space. A shared bank of directed cross-modal experts models ordered modality interactions; task-specific gates combine expert outputs before the emotion, personality, and ambivalence heads. The pipeline highlights five components: statistical aggregation, directed experts, task-aware expert mixtures, FM regularization, and PAGB.

4.2 Modality Encoding and Aggregation

For each task t , example i , and modality $m \in \mathcal{M}$, a feature extractor Φ_m maps the raw modality stream

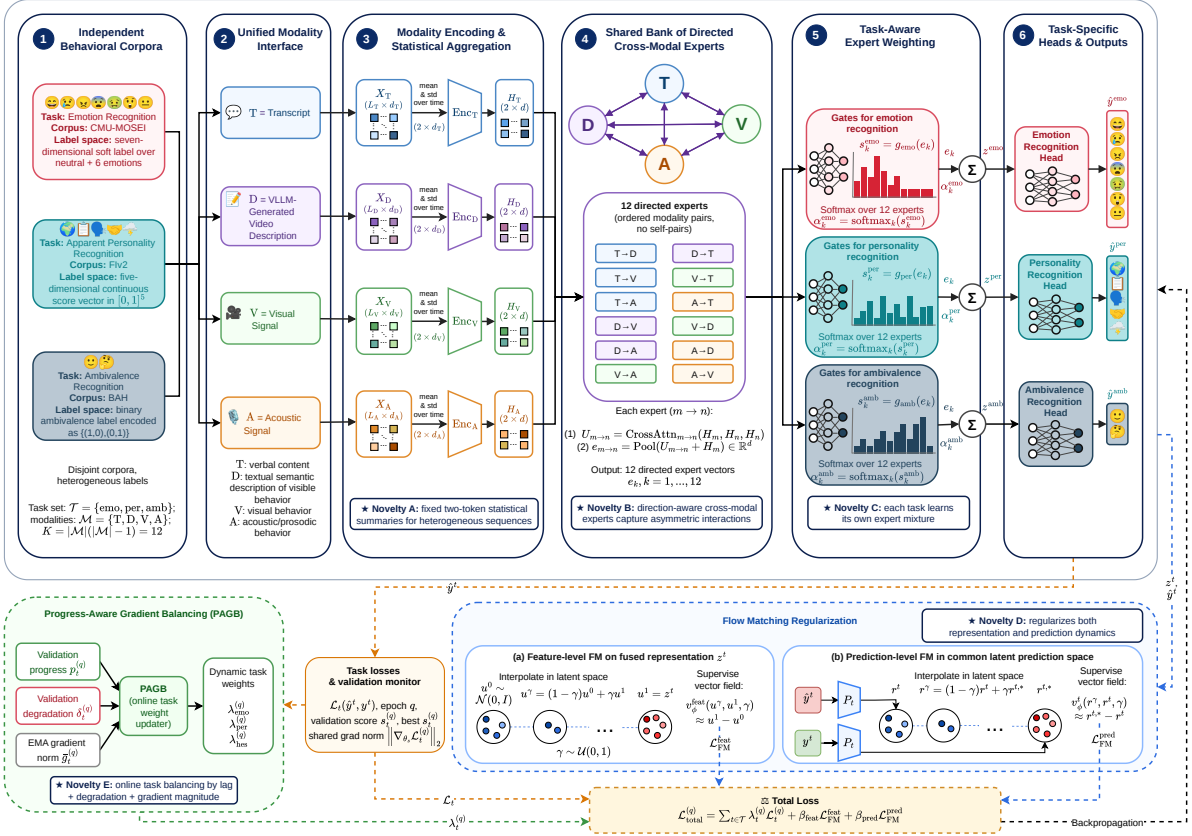


Figure 1: Overview of TACME. Independent corpora are mapped to a unified modality interface, encoded through two-token statistical summaries, processed by directed cross-modal experts, and combined through task-aware gates. The lower part shows feature- and prediction-level FM and the PAGB training controller.

$x_{i,m}^t$ to a feature matrix

$$\bar{X}_{i,m}^t = \Phi_m(x_{i,m}^t) \in \mathbb{R}^{L_{i,m}^t \times d_m}, \quad (6)$$

where $L_{i,m}^t$ is the number of temporal positions, tokens, or frame-level units, and d_m is the feature dimension for modality m .

Since clips differ in duration across independently collected corpora, the modality feature matrices also differ in length. We therefore summarize each feature matrix by its mean and STD over the sequence dimension:

$$\bar{X}_{i,m}^t = \begin{bmatrix} \mu(X_{i,m}^t) \\ \sigma(X_{i,m}^t) \end{bmatrix} \in \mathbb{R}^{2 \times d_m}, \quad (7)$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ are computed over the first dimension of $X_{i,m}^t$, i.e., over temporal positions, tokens, or frames. The bracket notation denotes vertical stacking of the mean and STD vectors. Thus, variable-length modality streams are converted into fixed-length two-token statistical summaries.

Because different modalities can have different feature dimensions d_m , each statistical summary is

then mapped to a shared hidden dimension d by a modality-specific encoder:

$$H_{i,m}^t = \text{Enc}_m(\bar{X}_{i,m}^t) \in \mathbb{R}^{2 \times d}. \quad (8)$$

The resulting representation contains two tokens (mean and STD) with a common hidden size d , which allows the subsequent Transformer-based cross-modal experts to operate on all modalities in a shared space. To make the formal description easier to follow, we summarize the main notation used in the problem formulation and method in Appendix D.

4.3 Directed Cross-Modal Experts

For two modalities $m, n \in \mathcal{M}$, $m \neq n$, a directed expert $E_{m \rightarrow n}$ models how modality m attends to modality n . We use $\text{CrossAttn}_{m \rightarrow n}$ to denote a standard multi-head cross-attention block whose parameters are specific to the ordered modality pair $m \rightarrow n$. It takes a query sequence, a key sequence, and a value sequence, and returns a sequence with the same length as the query. We use $H_{i,m}^t$ as the

query sequence and $H_{i,n}^t$ as both key and value:

$$U_{i,m \rightarrow n}^t = \text{CrossAttn}_{m \rightarrow n}(H_{i,m}^t, H_{i,n}^t, H_{i,n}^t), \quad (9)$$

where $U_{i,m \rightarrow n}^t \in \mathbb{R}^{2 \times d}$. The expert output is obtained by residual pooling:

$$e_{i,m \rightarrow n}^t = \text{Pool}(U_{i,m \rightarrow n}^t + H_{i,m}^t) \in \mathbb{R}^d. \quad (10)$$

$E_{m \rightarrow n}$ and $E_{n \rightarrow m}$ use different query modalities and therefore capture different cross-modal relations. With four modalities and no self-pairs, the bank contains $K = |\mathcal{M}|(|\mathcal{M}| - 1) = 12$ experts. We write the expert outputs as $\mathcal{E}(x_i^t) = \{e_{i,k}^t\}_{k=1}^K$, where each k corresponds to one ordered modality pair.

4.4 Task-Aware Expert Weighting

Given the expert outputs $\mathcal{E}(x_i^t)$, each task t uses a separate gate to assign weights to the shared expert bank. For expert k , the task-specific gate first computes a scalar score:

$$s_{i,k}^t = g_t(e_{i,k}^t), \quad (11)$$

where $g_t : \mathbb{R}^d \rightarrow \mathbb{R}$ is a scoring network for task t . The normalized expert weight is then

$$\alpha_{i,k}^t = \frac{\exp(s_{i,k}^t)}{\sum_{r=1}^K \exp(s_{i,r}^t)}. \quad (12)$$

Thus, $\alpha_{i,k}^t \geq 0$ and $\sum_{k=1}^K \alpha_{i,k}^t = 1$. The task-specific fused representation is

$$z_i^t = \sum_{k=1}^K \alpha_{i,k}^t e_{i,k}^t \in \mathbb{R}^d. \quad (13)$$

The final prediction for task t is produced by a task-specific head:

$$\hat{y}_i^t = h_t(z_i^t), \quad (14)$$

where h_t maps the fused representation to the target space $\mathcal{Y}_t$.

For a controlled comparison, we also evaluate an equal-expert variant that removes task-aware gating:

$$z_{i,\text{eq}}^t = \frac{1}{K} \sum_{k=1}^K e_{i,k}^t. \quad (15)$$

The comparison between Eq. 13 and Eq. 15 tests whether task-dependent expert weighting is useful.

4.5 Flow Matching Regularization

We evaluate FM (Jha et al., 2026) as an additional regularization mechanism at two levels: fused representations and task predictions. At the feature level, we apply FM to the task-specific fused representation z_i^t . Let $u_{i,t}^0 \sim \mathcal{N}(0, I)$ be a noise vector with the same dimension as z_i^t , and let $u_{i,t}^1 = z_i^t$. For $\gamma \sim \mathcal{U}(0, 1)$, we define the interpolation

$$u_{i,t}^\gamma = (1 - \gamma)u_{i,t}^0 + \gamma u_{i,t}^1. \quad (16)$$

In implementation, the expectation over γ is approximated by averaging over S_{FM} sampled interpolation points. The feature-level FM loss is

$$\mathcal{L}_{\text{FM}}^{\text{feat}} = \mathbb{E} \left[\left\| v_\phi^{\text{feat}}(u_{i,t}^\gamma, z_i^t, \gamma) - (u_{i,t}^1 - u_{i,t}^0) \right\|_2^2 \right], \quad (17)$$

where v_ϕ^{feat} is a learnable vector field, $\|\cdot\|_2^2$ denotes the squared Euclidean norm, and $\mathbb{E}[\cdot]$ denotes the expectation over tasks, examples, noise samples, and interpolation values.

For prediction-level FM, we avoid comparing logits from heterogeneous task spaces directly. Instead, each task has a projection P_t that maps predictions and targets to a common latent prediction space:

$$r_i^t = P_t(\hat{y}_i^t), \quad r_i^{t,*} = P_t(y_i^t). \quad (18)$$

Here, $\hat{y}_i^t = h_t(z_i^t)$ is the task prediction and y_i^t is the corresponding target. We then define

$$r_{i,t}^\gamma = (1 - \gamma)r_i^t + \gamma r_i^{t,*}, \quad (19)$$

and train a task-specific vector field:

$$\mathcal{L}_{\text{FM}}^{\text{pred}} = \mathbb{E} \left[\left\| v_\psi^t(r_{i,t}^\gamma, z_i^t, \gamma) - (r_i^{t,*} - r_i^t) \right\|_2^2 \right]. \quad (20)$$

The total training loss at step or epoch q is

$$\mathcal{L}_{\text{total}}^{(q)} = \sum_{t \in \mathcal{T}} \lambda_t^{(q)} \mathcal{L}_t^{(q)} + \beta_{\text{feat}} \mathcal{L}_{\text{FM}}^{\text{feat}} + \beta_{\text{pred}} \mathcal{L}_{\text{FM}}^{\text{pred}}, \quad (21)$$

where $\mathcal{L}_t^{(q)}$ is the task loss at step or epoch q , and β_{feat} and β_{pred} control the contributions of feature- and prediction-level FM.

4.6 Progress-Aware Gradient Balancing

We also evaluate PAGB, which updates task weights using three online signals: relative task progress, recent validation degradation, and smoothed gradient magnitude. All task validation

scores are oriented so that higher values indicate better performance.

Let $s_t^{(q)}$ be the validation score of task t after epoch q , and let $s_t^{(0)}$ be its initial validation score. We denote the best score observed up to epoch q as

$$s_t^*(q) = \max_{0 \leq r \leq q} s_t^{(r)}. \quad (22)$$

The relative progress and degradation terms are defined as

$$p_t^{(q)} = \frac{s_t^{(q)}}{\max(s_t^{(0)}, \epsilon)}, \quad (23)$$

$$\delta_t^{(q)} = \max \left(0, \frac{s_t^*(q) - s_t^{(q)}}{|s_t^*(q)| + \epsilon} \right), \quad (24)$$

where $\epsilon > 0$ is a small constant for numerical stability. The progress term $p_t^{(q)}$ measures the current validation score relative to the initial validation score, while $\delta_t^{(q)}$ measures how far the current score is below the best score observed so far.

We define the task lag as

$$\ell_t^{(q)} = \max \left(0, \max_{r \in \mathcal{T}} p_r^{(q)} - p_t^{(q)} \right) + \delta_t^{(q)}. \quad (25)$$

Thus, a task receives a larger lag when it improves more slowly than the fastest-progressing task or when its validation score has recently degraded.

Let $\theta_s \subset \theta$ be the shared model parameters. For task t , we compute the gradient norm

$$g_t^{(q)} = \left\| \nabla_{\theta_s} \mathcal{L}_t^{(q)} \right\|_2, \quad (26)$$

where $\mathcal{L}_t^{(q)}$ is the task loss at epoch q . PAGB maintains an Exponential Moving Average (EMA) of this norm:

$$\bar{g}_t^{(q)} = \eta_g \bar{g}_t^{(q-1)} + (1 - \eta_g) g_t^{(q)}. \quad (27)$$

where $\eta_g \in [0, 1)$ is the gradient-norm smoothing coefficient.

The raw task weight is then computed as

$$\tilde{\lambda}_t^{(q)} = \frac{\exp(\kappa_p \ell_t^{(q)})}{\left(\frac{\bar{g}_t^{(q)}}{\frac{1}{|\mathcal{T}|} \sum_{r \in \mathcal{T}} \bar{g}_r^{(q)} + \epsilon} \right)^{\gamma_p}}, \quad (28)$$

where $\kappa_p \geq 0$ controls the sensitivity to task lag, and $\gamma_p \geq 0$ controls the strength of gradient-norm correction. Finally, raw weights are normalized as

$$\lambda_t^{(q)} = \frac{|\mathcal{T}| \tilde{\lambda}_t^{(q)}}{\sum_{r \in \mathcal{T}} \tilde{\lambda}_r^{(q)}}, \quad (29)$$

so that the average task weight remains one. The normalized weights are clipped to $[\lambda_{\min}, \lambda_{\max}]$ and smoothed with coefficient η_λ before being used in the shared update. This final step prevents abrupt changes in task weights during shared-parameter updates.

5 Experimental Setup

5.1 Research Corpora

We use three independently collected corpora, each associated with a different behavioral task. Because no corpus contains annotations for all target tasks, the resulting setting is disjoint by construction. General information about the corpora is presented in Appendix E.

CMU-MOSEI (Zadeh et al., 2018) provides short monologue segments annotated with six Ekman emotion-intensity scores. We convert these annotations into seven-dimensional soft labels over neutral and the six emotions for training; details are given in Appendix E.

Fiv2 (Escalante et al., 2022) targets apparent personality assessment from short videos. The labels correspond to the Big Five personality traits: openness (OPE), conscientiousness (CON), extraversion (EXT), agreeableness (AGR), and non-neuroticism (NNEU).

BAH (González-González et al., 2026) focuses on ambivalence recognition in behavior-change scenarios. It contains participant responses to structured questions and provides binary annotations for the presence of ambivalence.

5.2 Metrics and Model Selection

To evaluate the performance of the models, we use standard metrics for the selected research corpora (see Appendix E). Since the evaluation metrics are not directly comparable across tasks, we report all task metrics separately and use average rank only for model selection and aggregate comparison.

For each metric $r \in \mathcal{R}$, we first rank all compared models and denote the rank of model a by $\text{rank}_r(a)$. The average rank is then

$$\text{Rank}_{\text{avg}}(a) = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \text{rank}_r(a), \quad (30)$$

where lower values indicate a better cross-task trade-off. Checkpoints are selected by the same aggregate rank on the validation set. We use ranks only as an aggregate selection criterion and not as a replacement for task-specific metrics.

Model	Aggregation	FM setting	MTL strategy	CMU-MOSEI		Flv2		BAH		Avg. Rank ↓
				mMF1 ↑	mWAcc ↑	mACC ↑	CCC ↑	MF1 ↑	UAR ↑	
UniGraph2 (He et al., 2025)	Mean	W/o FM	Equal	59.42	60.29	90.62	54.42	67.38	68.89	14.17
UniGraph2	Mean	FL and PL	Equal	60.41	61.15	91.23	61.53	68.90	68.97	11.17
UniGraph2	Mean	FL and PL	PAGB	61.31	61.61	91.25	66.54	68.70	69.36	9.50
DPDF-LQ (Zhou et al., 2025a)	Mean + STD	W/o FM	Equal	62.05	62.76	90.99	57.95	68.00	69.18	9.67
DPDF-LQ	Mean + STD	FL and PL	Equal	62.45	62.89	90.90	58.15	68.84	69.71	7.17
DPDF-LQ	Mean + STD	FL and PL	PAGB	62.38	62.73	91.04	62.77	69.44	70.03	5.33
HME (Zhuang et al., 2025)	Mean	W/o FM	Equal	61.92	62.59	91.51	64.63	66.62	66.95	9.83
HME	Mean	FL	Equal	62.10	62.30	91.47	65.08	67.64	69.38	8.17
HME	Mean	FL	PAGB	62.28	62.30	91.62	64.89	68.16	67.99	8.17
ComP (He et al., 2026)	Mean + STD	W/o FM	Equal	59.17	61.56	90.52	50.82	69.32	69.47	12.00
ComP	Mean + STD	FL and PL	Equal	60.56	61.81	90.48	51.72	68.90	69.96	11.50
ComP	Mean + STD	FL and PL	PAGB	60.71	61.93	90.49	51.95	69.38	68.87	11.50
TACME with equal experts	Mean + STD	W/o FM	Equal	61.76	62.50	91.28	64.92	69.06	70.02	6.50
TACME with task-aware experts	Mean + STD	W/o FM	Equal	61.88	62.45	91.58	67.73	69.05	70.01	5.83
TACME with task-aware experts	Mean + STD	FL and PL	Equal	62.25	62.80	91.60	68.51	68.74	70.37	4.17
TACME with task-aware experts	Mean + STD	FL and PL	PAGB	62.68	63.29	91.71	69.89	71.15	72.07	1.00

Table 1: Comparison of base multimodal fusion models and their variants with FM regularization and the proposed PAGB MTL strategy. FL and PL denote feature-level and prediction-level FM, respectively. **Best** and **second-best** results are highlighted.

Strategy	CMU-MOSEI		Flv2		BAH		Avg. Rank ↓
	mMF1 ↑	mWAcc ↑	mACC ↑	CCC ↑	MF1 ↑	UAR ↑	
Equal weighting	62.25	62.80	91.60	68.51	68.74	70.37	6.67
Learned weighting	61.93	62.62	91.62	68.91	69.66	71.03	5.83
NTKMTL (Qin et al., 2025)	62.56	62.92	91.90	71.45	67.97	69.44	4.67
FairGrad (Ban and Ji, 2024)	62.59	63.17	91.90	72.44	67.22	68.07	3.83
PCGrad (Yu et al., 2020)	62.55	63.08	91.61	69.10	69.06	70.65	5.00
CAGrad (Liu et al., 2021)	62.66	63.17	91.86	71.29	70.04	71.49	2.67
IMGrad (Zhou et al., 2025b)	62.40	62.96	91.90	71.48	68.35	69.90	4.33
PAGB (ours)	62.68	63.29	91.71	69.89	71.15	72.07	2.33

Table 2: Comparison of MTL strategies. All variants use task-aware TACME with Mean + STD aggregation and the feature- and prediction-level FM setting. **Best** and **second-best** results are highlighted.

5.3 Comparison Protocol

We focus on intermediate feature-level fusion rather than early or late fusion, following prior multimodal studies that report stronger performance for this setting (Zhou et al., 2025a; He et al., 2026). We evaluate seven groups of experiments: (i) TACME with equal and task-specific expert weighting; (ii) feature-level, prediction-level, and combined FM; (iii) alternative fusion models, including UniGraph2 (He et al., 2025), DPDF-LQ (Zhou et al., 2025a), HME (Zhuang et al., 2025), and ComP (He et al., 2026); (iv) aggregation strategies: mean pooling, mean and STD summaries, and raw; (v) MTL strategies: equal weighting, learned weighting, NTKMTL (Qin et al., 2025), FairGrad (Ban and Ji, 2024), PCGrad (Yu et al., 2020), CAGrad (Liu et al., 2021), IMGrad (Zhou et al., 2025b), and PAGB; (vi) modality ablations and single-task comparisons; and (vii) the trade-off between single-task and multi-task TACME using zero-shot Multimodal EmotionLines Dataset (MELD) transfer and corpus-identification diagnostics. We used exclusively open source solutions as fusion models and MTL strategies.

All models use the data splits provided by the corpus authors and the same feature extraction protocol. Model-specific and training hyperparameters are listed in Appendix F; computational cost is reported in Appendix G. The VLLM prompt is provided in Appendix H, and representative generated descriptions with their audits are shown in Appendix I. Multi-task TACME results across demographic subgroups are provided in Appendix J. Tables 1 and 2 are controlled comparisons within this shared pipeline, Table 3 is an internal modality ablation, and Table 4 compares TACME with literature-reported task-specific systems.

6 Results and Analysis

6.1 Comparison of Multimodal Fusion Models

Table 1 compares TACME with recent multimodal fusion models under the same disjoint-corpus MTL protocol. Both TACME variants outperform the adapted baselines in average rank; even equal experts give a lower average rank than all baseline models, supporting the value of directed modality-pair interactions. Task-aware expert weighting further improves the rank, confirming that differ-

Variant	CMU-MOSEI		Flv2		BAH		Avg. Rank ↓
	mMF1 ↑	mWAcc ↑	mACC ↑	CCC ↑	MF1 ↑	UAR ↑	
Full model	62.68	63.29	91.71	69.89	71.15	72.07	2.17
w/o transcript	60.68	61.84	91.66	69.75	65.78	66.58	6.00
w/o VLLM-generated behavioral descriptions	62.92	63.44	91.44	69.27	68.81	69.92	3.50
w/o visual signal	60.80	61.69	90.32	52.17	73.91	74.23	5.17
w/o acoustic signal	61.64	61.96	91.45	67.98	67.25	68.70	5.00
w/o acoustic signal + transcript	57.06	58.35	91.14	64.10	44.61	45.52	9.67
w/o acoustic signal + VLLM-generated behavioral descriptions	61.47	61.77	91.26	65.57	69.00	70.46	4.83
w/o acoustic signal + visual signal	59.68	60.98	89.50	38.95	67.14	67.88	9.67
w/o transcript + VLLM-generated behavioral descriptions	60.54	61.29	91.76	70.36	68.01	68.01	5.33
w/o transcript + visual signal	59.00	60.52	90.41	53.95	68.55	69.82	8.00
w/o VLLM-generated behavioral descriptions + visual signal	60.21	61.31	90.15	50.83	71.08	72.52	6.67

Table 3: Modality ablation of the final model. All variants use task-aware TACME with Mean + STD aggregation and the feature- and prediction-level FM setting, and PAGB. **Smallest** and **largest** degradations are highlighted.

CMU-MOSEI			Flv2			BAH		
Method	mMF1 ↑	mWAcc ↑	Method	mACC ↑	CCC ↑	Method	MF1 ↑	UAR ↑
MASAI (Markitantov et al., 2026)	–	55.90	DLF (Ning et al., 2026)	91.77	–	DivFusion (Souza et al., 2026)	68.08	–
Arumugam et al. (2025)	–	62.10	MoRE (Zhang et al., 2025)	91.99	–	ConflictAwareAH (Bekhouché et al., 2026)	69.40	–
MER-CAPE (Chen et al., 2026)	66.10	67.38	MP-PMGRT (Wang et al., 2025b)	92.07	–	BROTHER (Pereira et al., 2026)	74.65	–
Single-task TACME (ours)	67.82 ± 0.37	68.17 ± 0.42	Single-task TACME (ours)	92.59 ± 0.18	77.47 ± 1.17	Single-task TACME (ours)	78.05 ± 0.62	78.63 ± 0.78
Multi-task TACME (ours)	62.54 ± 0.26	63.32 ± 0.24	Multi-task TACME (ours)	91.76 ± 0.05	70.14 ± 0.78	Multi-task TACME (ours)	70.29 ± 0.90	71.71 ± 0.53
Δ (multi-task vs. single-task)	−5.28	−4.85	Δ (multi-task vs. single-task)	−0.83	−7.33	Δ (multi-task vs. single-task)	−7.76	−6.92

Table 4: Comparison with task-specific State-of-the-Art (SOTA) systems. The reported SOTA values use corpus protocols comparable to those used for TACME. For TACME, results are reported as mean ± STD over five runs with different random seeds; the final row reports the multi-task minus single-task difference.

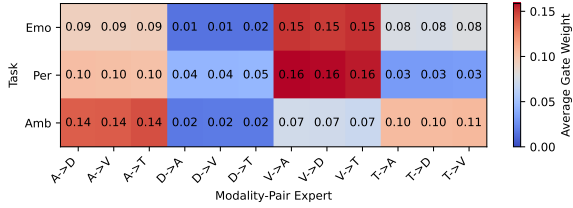


Figure 2: Average task-specific gate weights $\alpha_{i,k}^t$ assigned to directed cross-modal experts $E_{m \rightarrow n}$. Weights are averaged over test samples.

ent tasks require different cross-modal evidence. With FM and PAGB, task-aware TACME achieves the best overall trade-off with the lowest average rank. Figure 2 illustrates this behavior: emotion and personality mainly select visual-query experts, whereas ambivalence shifts toward acoustic- and transcript-query experts. Thus, the gates learn task-dependent directed interactions instead of one shared fusion pattern. Additional analyses of aggregation, FM, and input attention are provided in Appendices K, L, and M.

6.2 Multi-Task Learning Strategies

Table 2 compares MTL strategies using the same task-aware TACME configuration. Equal weighting gives the weakest average rank, while learned weighting gives only a modest improvement because it does not use task progress or gradient dynamics. Gradient-based methods are stronger, with

CAGrad achieving the best baseline rank of 2.67. PAGB obtains the best overall rank of 2.33 and the highest CMU-MOSEI and BAH scores, indicating that progress-aware task weighting better balances unstable behavioral tasks. FairGrad and NTKMTL improve Flv2 CCC, but these gains coincide with weaker BAH performance, whereas PAGB gives the most balanced cross-task trade-off. Additional convergence curves, PAGB component ablations, and training-dynamics visualizations are provided in Appendices N, O, and P.

6.3 Modality Ablation

Table 3 shows that the full four-stream model achieves the lowest average rank of 2.17. Removing any single modality worsens the aggregate rank, confirming that transcript, VLLM-generated behavioral descriptions, visual signal, and acoustic signal provide complementary cues. The largest single-modality drop occurs without transcript, especially on BAH, indicating the importance of verbal content for ambivalence recognition. Removing the acoustic signal hurts ambivalence and emotion recognition, while removing the visual signal degrades personality recognition.

Double-modality ablations produce stronger failures: removing both the acoustic signal and transcript gives the worst BAH scores, whereas removing both acoustic and visual signals severely hurts Flv2. Thus, TACME benefits from complementary

Feature source	MF1 ↓	WF1 ↓	UAR ↓	Accuracy ↓
Verbal content features	92.97	95.19	91.37	95.22
Textual visible-behavior description features	90.20	93.21	91.12	93.14
Visual behavior features	87.64	89.77	86.49	89.63
Acoustic/prosodic behavior features	95.82	95.84	95.94	95.82
Multi-task TACME: target-corpus expert features	96.28	97.69	96.02	97.70
Single-task TACME: target-corpus expert features	100.00	100.00	100.00	100.00
Multi-task TACME: non-target-corpus expert features	92.85	93.73	91.87	93.74
Single-task TACME: non-target-corpus expert features	95.34	96.39	94.52	96.39

Table 5: Corpus-identification results. Higher values indicate stronger corpus separability.

verbal, acoustic, visual, and behavior-description streams rather than relying on one dominant modality. Additional prediction-level task correlation analysis is provided in Appendix Q.

6.4 Single-Task Comparison

Table 4 compares TACME with literature-reported task-specific systems for each corpus. The reported SOTA results are taken from cited studies using comparable corpus protocols. For TACME, we report mean and STD over five random seeds; detailed results are provided in Appendix R.

Single-task TACME outperforms the compared task-specific SOTA systems on the reported metrics. This gain is consistent with the proposed directed cross-modal expert architecture, which shares modality-pair Transformer experts while selecting task-relevant cross-modal interactions through task-specific gates. The relatively small STDs across most metrics indicate stable performance across random seeds.

However, single-task TACME also outperforms multi-task TACME on all six source-corpus metrics. This reflects the cost of joint optimization across disjoint corpora with different populations, recording conditions, annotation protocols, and modality relevance. Thus, single-task TACME remains preferable when performance on an individual corpus is the primary objective.

6.5 Corpus-Identification Probe

We examine corpus specialization by training a one-layer linear classifier to predict corpus identity (CMU-MOSEI = 0, Fiv2 = 1, BAH = 2) from frozen modality and task-aware expert features. For TACME, expert features are extracted from separately trained single-task models and from the multi-task model. Higher scores indicate stronger corpus separability and are therefore undesirable.

Corpus identity is highly predictable from all pretrained modality features, confirming substantial differences among the corpora. However, corpus-identification scores are consistently lower

Model	Recall $\uparrow$							UAR $\uparrow$
	Neutral	Anger	Disgust	Fear	Happiness	Sadness	Surprise	
Single-task TACME	33.1	6.4	16.2	0.0	91.3	24.5	31.0	28.9
Multi-task TACME	36.1	7.5	50.0	6.0	65.7	36.5	49.8	35.9

Table 6: Zero-shot class-wise recall on unseen MELD.

for multi-task than for single-task TACME expert features, indicating that multi-task training reduces, but does not eliminate, corpus specialization.

6.6 Zero-Shot Generalization on MELD

To evaluate generalization to unseen data, we compare single-task and multi-task TACME on the MELD test set (Poria et al., 2019). MELD is not used for training or model selection and contains 2,610 test utterances from seven emotion classes.

Table 6 shows that multi-task TACME improves Unweighted Average Recall (UAR) from 28.9% to 35.9%, with substantial gains for disgust, sadness, and surprise, although happiness recall decreases. Overall, these results indicate that multi-task TACME trades some source-corpus performance for reduced corpus specialization and improved zero-shot generalization.

Equivalent zero-shot evaluation is unavailable for the other tasks: BAH is the only suitable ambivalence corpus, while available personality corpora use self-reported rather than observer-rated apparent personality traits.

7 Conclusion

This work addressed disjoint-corpus multimodal multi-task behavioral understanding, where emotion, personality, and ambivalence recognition are learned jointly from independent corpora. TACME introduces task-aware directed cross-modal experts with task-specific gates, complemented by two-token statistical aggregation, feature- and prediction-level FM, and PAGB. Across three corpora and six task-specific metrics, the final multi-task TACME configuration achieves the best average rank of 1.00 among the compared multimodal fusion models. Single-task TACME outperforms the compared task-specific SOTA systems and achieves higher source-corpus performance than multi-task TACME. In contrast, multi-task TACME shows lower corpus specialization and improves zero-shot UAR on unseen MELD from 28.9% to 35.9%, indicating a trade-off between source-corpus performance and generalization to unseen data. The source code is publicly available at <https://github.com/LEYA-HSE/TACME>.

Limitations

This work has several limitations. First, because the corpora are disjoint, the model cannot be evaluated on all tasks for the same individuals, and task identity remains coupled with corpus identity. The corpus-identification probe shows that corpus separability remains high, so corpus-task confounding is not eliminated. Second, despite the neutral prompt, VLLM-generated descriptions may contain interpretive or demographic references; Appendix I reports an explicit audit. Third, statistical aggregation reduces computational cost but may discard fine-grained temporal information. Fourth, the directed expert bank grows as $|\mathcal{M}|(|\mathcal{M}| - 1)$, so larger modality sets may require sparse expert selection, shared parameterization, or pruning. Finally, TACME assumes that the tasks share some behavioral structure. Weakly related tasks may still cause negative transfer, and generalization to domains with different objectives or modality relationships requires additional validation.

Ethical Considerations

We use available research corpora for model training and evaluation, following their intended research use and data-use terms. The proposed system addresses behavioral analysis, including emotion, apparent personality, and ambivalence recognition. Such systems should be used with informed consent and clear disclosure. Predictions of personality or ambivalence should not be treated as objective psychological diagnoses or as grounds for high-stakes individual decisions without human oversight. We recommend auditing model behavior across demographic, linguistic, and cultural groups. VLLM-generated descriptions should be prompted to describe observable behavior rather than infer mental states or target labels. Appendix I separately audits generated descriptions for target-label leakage, mental-state inference, and demographic/stereotypical language. Appendix J reports diagnostic subgroup results for Flv2 and BAH; performance varies across demographic groups, and these results should not be interpreted as a definitive fairness assessment.

Acknowledgements

This work was supported by the grant for research centers in the field of Artificial Intelligence (AI) provided by the Ministry of Economic Development of the Russian Federation in accordance with

Agreement No. 000000C313925P4E0002 and the agreement with HSE University No. 139-15-2025-009, with partial support from the computational resources of High-Performance Computing (HPC) facilities at HSE University.

References

- Shahin Amiriparian, Filip Packań, Maurice Gerczuk, and Björn W Schuller. 2024. [ExHuBERT: Enhancing HuBERT through block extension and fine-tuning on 37 emotion datasets](#). In *Interspeech*, pages 2635–2639.
- Lalitha Arumugam, Samyudurai Arumugam, Pabitha Chidambaram, and Kumaresan Govindasamy. 2025. [A multi-model deep learning approach for human emotion recognition](#). *Cognitive Neurodynamics*, 19(1):123.
- Hao Ban and Kaiyi Ji. 2024. [Fair resource allocation in multi-task learning](#). In *International Conference on Machine Learning (ICML)*, pages 2715–2731.
- Salah Eddine Bekhouche, Hichem Telli, Azeddine Benlamoudi, Salah Eddine Herrouz, Abdelmalik Taleb-Ahmed, and Abdenour Hadid. 2026. [Conflict-aware multimodal fusion for ambivalence and hesitancy recognition](#). *ArXiv*.
- Thomas Borsani, Andrea Rosani, Giuseppe Nicosia, and Giuseppe Di Fatta. 2025. [Gradient similarity surgery in multi-task deep learning](#). In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 95–111.
- Chengming Chen, Pengyuan Liu, Zhicheng Dong, Zhuo He, and Zhijian Li. 2026. [MER-CAPF: Audio-text emotion recognition through cross-attention mechanism and multi-granularity pooling strategy](#). *Pattern Recognition Letters*, 201:125–131.
- Hugo Jair Escalante, Heysem Kaya, Albert Salah, Sergio Escalera, Yagmur Gucluturk, Umut Guclu, Xavier Baró, Isabelle Guyon, Julio Jacques Junior, Meysam Madadi, Stephane Ayache, Evelyn Viegas, Furkan Gürpınar, Achmadnoer Wicaksana, Cynthia Liem, Marcel Van Gerven, and Rob Van Lier. 2022. [Modeling, recognizing, and explaining apparent personality from videos](#). *IEEE Transactions on Affective Computing*, 13(2):894–911.
- Kshitish Ghate, Tessa Charlesworth, Mona Diab, and Aylin Caliskan. 2025. [Biases propagate in encoder-based vision-language models: A systematic analysis from intrinsic measures to zero-shot retrieval outcomes](#). In *Findings of the Association for Computational Linguistics: ACL*, pages 18562–18580.
- Manuela González-González, Soufiane Belharbi, Muhammad Zeeshan, Masoumeh Sharafi, Muhammad Haseeb Aslam, Marco Pedersoli, Alessandro Lameiras Koerich, Simon L Bacon, and Eric Granger.

2026. [BAH dataset for ambivalence/hesitancy recognition in videos for digital behavioural change](#). In *International Conference on Learning Representations (ICLR)*, pages 41540–41586.
- Wen-Jue He, Xiaofeng Zhu, and Zheng Zhang. 2026. [Cross-modal prompting for balanced incomplete multi-modal emotion recognition](#). In *AAAI Conference on Artificial Intelligence*, volume 40, pages 17463–17471.
- Yufei He, Yuan Sui, Xiaoxin He, Yue Liu, Yifei Sun, and Bryan Hooi. 2025. [Unigraph2: Learning a unified embedding space to bind multimodal graphs](#). In *ACM Web Conference (WWW)*, pages 1759–1770.
- Om Govind Jha, Manoj Bamniya, and Ayon Borthakur. 2026. [Discriminative flow matching via local generative predictors](#). *ArXiv*.
- Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. 2021. [Conflict-averse gradient descent for multi-task learning](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 18878–18890.
- Ziyang Ma, Mingjie Chen, Hezhao Zhang, Zhisheng Zheng, Wenxi Chen, Xiquan Li, Jiaxin Ye, Xie Chen, and Thomas Hain. 2024. [EmoBox: Multilingual multi-corpus speech emotion recognition toolkit and benchmark](#). In *Interspeech*, pages 1580–1584.
- Maxim Markitantov, Elena Ryumina, Anastasia Dvoynikova, and Alexey Karpov. 2026. [Multilingual approach for multi-modal emotion and sentiment recognition based on triple fusion](#). *Information Fusion*, 132:104207.
- Tao Ning, ZhengHua Guo, and QiDong Hou. 2026. [A DLF multi-scale quantitative research method for the big five personality traits](#). *Scientific Reports*, 16:251.
- Alexandre Pereira, Pablo Barros, and Bruno Fernandes. 2026. [BROTHER: Behavioral recognition optimized through heterogeneous ensemble regularization for ambivalence and hesitancy](#). In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 5362–5369.
- Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. 2019. [MELD: A multimodal multi-party dataset for emotion recognition in conversations](#). In *Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 527–536.
- Xiaohan Qin, Xiaoxing Wang, Ning Liao, and Junchi Yan. 2025. [NTKMTL: Mitigating task imbalance in multi-task learning from neural tangent kernel perspective](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, pages 53856–53884.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *International Conference on Machine Learning (ICML)*, pages 8748–8763.
- Elena Ryumina, Alexandr Axyonov, Darya Koryakovskaya, Timur Abdulkadirov, Angelina Egorova, Sergey Fedchin, Alexander Zaburdaev, Iana Salova, Andrey Savchenko, and Dmitry Ryumin. 2026. [MM-PSYCHE: Multimodal multitask psychological characteristic estimation through cross-domain semi-supervised learning](#). *IEEE Access*, 14:124759–124778.
- Islam JAM Samiul, Khalid Zaman, Marius Funk, Masashi Unoki, Yukiko Nakano, and Shogo Okada. 2025. [Culture-aware multimodal personality prediction using audio, pose, and cultural embeddings](#). In *International Conference on Multimodal Interaction Companion (ICMI Companion)*, pages 222–226.
- Jivnesh Sandhan, Fei Cheng, Tushar Sandhan, and Yugo Murawaki. 2025. [CAPE: Context-aware personality evaluation framework for large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP*, pages 10648–10662.
- Aislan Gabriel O Souza, Bruno Fernandes, Leandro Honorato Silva, and Agostinho Freire. 2026. [Divergence-based multimodal fusion for ambivalence and hesitancy recognition](#). In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 5446–5452.
- Johannes Wagner, Andreas Triantafyllopoulos, Hagen Wierstorf, Maximilian Schmitt, Felix Burkhardt, Florian Eyben, and Björn W Schuller. 2023. [Dawn of the transformer era in speech emotion recognition: closing the valence gap](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(9):10745–10759.
- Fanfan Wang, Xiangqing Shen, Jianfei Yu, and Rui Xia. 2025a. [Flexible thinking for multimodal emotional support conversation via reinforcement learning](#). In *Findings of the Association for Computational Linguistics: EMNLP*, pages 1341–1356.
- Rongquan Wang, Xianyu Xu, Hao Yang, Lin Wei, and Huimin Ma. 2025b. [A novel multimodal personality prediction method based on pretrained models and graph relational transformer network](#). In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5.
- Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. 2020. [Gradient surgery for multi-task learning](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 5824–5836.
- AmirAli Bagher Zadeh, Paul Pu Liang, Soujanya Poria, Erik Cambria, and Louis-Philippe Morency. 2018. [Multimodal language analysis in the wild: CMU-MOSEI dataset and interpretable dynamic fusion graph](#). In *Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 2236–2246.

Huanzhen Zhang, Yuhang Li, Chengwei Ye, Yufei Lin, and Bohan Hu. 2025. [MoRE: Structured multisignal encoding for human disposition recognition from short media clips](#). In *Advanced Intelligent Computing Technology and Applications: International Conference (ICIC)*, pages 48–57.

Miao Zhou, Lina Yang, Thomas Wu, Dongnan Yang, and Xinru Zhang. 2025a. [Dual-path dynamic fusion with learnable query for multimodal sentiment analysis](#). In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 11355–11365.

Zhipeng Zhou, Liu Liu, Peilin Zhao, and Wei Gong. 2025b. [Injecting imbalance sensitivity for multi-task learning](#). In *International Joint Conference on Artificial Intelligence (IJCAI)*, pages 7236–7244.

Yan Zhuang, Minhao Liu, Wei Bai, Yanru Zhang, Wei Li, Jiawen Deng, and Fuji Ren. 2025. [Hypermodality enhancement for multimodal sentiment analysis with missing modalities](#). In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, pages 157679–157708.

A Role of the Proposed Components

This section clarifies the role of the proposed components highlighted in Figure 1. Table 7 summarizes their functions and contribution status. The main architectural contribution is the task-aware directed expert design. Two-token statistical aggregation, FM, and PAGB are complementary representation, regularization, and optimization components evaluated around this core.

Two-token statistical aggregation. Standard graph-style multimodal models often aggregate a modality into one vector, while temporal models process full token or frame sequences. TACME uses an intermediate representation: each modality stream is summarized by two tokens, one for the mean and one for the STD over the sequence dimension. This keeps a fixed-length interface across corpora and modalities while preserving both central tendency and dispersion. The aggregation results in Appendix K show that this representation improves the accuracy–runtime trade-off compared with mean-only summaries and raw temporal matrices.

Directed cross-modal experts. Many fusion models combine modalities through a single shared fusion block or through unordered modality interactions. In TACME, each expert corresponds to an ordered modality pair. This makes $m \rightarrow n$ different from $n \rightarrow m$, so the model can represent asymmetric relations such as acoustic cues attending to visible behavior and visible behavior attending to acoustic cues.

Task-aware expert weighting. The expert bank is shared, but the gate is task-specific. This differs from using one global multimodal representation for all tasks. It lets emotion, personality, and ambivalence recognition select different mixtures of the same directed modality relations.

FM for heterogeneous representations and outputs. The FM variants regularize both fused representations and task-specific predictions. Prediction-level FM is applied in a common latent prediction space, which avoids directly forcing incompatible label spaces into a single output geometry.

PAGB combines validation progress, validation degradation, and EMA-smoothed gradient norms to update task weights online. This differs from purely learned task weights, which do not use validation dynamics, and from gradient-surgery methods, which modify gradients without explicitly

Component	Role	Status
Task-aware directed experts	Directed cross-modal interactions and task-aware routing	Main technical contribution
Flow Matching	Representation regularization	Complementary training component
PAGB	Multi-task loss balancing	Complementary optimization component

Table 7: Role of the central TACME components.

monitoring task progress.

B Compared Multimodal Fusion Models and Optimization Strategies

This section briefly summarizes the multimodal fusion models and MTL strategies used in the comparison. All methods are adapted to the same feature extraction pipeline, task heads, and disjoint-corpus training protocol.

B.1 Multimodal Fusion Models

UniGraph2 learns a unified embedding space for multimodal graphs by combining modality-specific encoders, graph neural networks, and expert-based alignment (He et al., 2025). We include it as a graph-based representation-alignment baseline. Unlike TACME, UniGraph2 is not designed around task-specific gates over directed modality-pair experts.

DPDF-LQ proposes dual-path dynamic fusion with a learnable query for multimodal sentiment analysis (Zhou et al., 2025a). It is included as a strong structured fusion baseline that separates global cross-modal dependencies from local modality information. In contrast to TACME, it does not explicitly maintain a bank of ordered modality-pair experts with task-specific weighting.

HME addresses missing-modality multimodal sentiment analysis through hyper-modality enhancement and uncertainty-aware fusion (Zhuang et al., 2025). We include it to test robustness under uneven modality reliability. Its main focus is sample-level modality incompleteness, whereas our setting also involves task-level heterogeneity across independent corpora.

Comp uses cross-modal prompting for balanced incomplete multimodal emotion recognition (He et al., 2026). It is relevant because it models cross-modal compensation under incomplete modalities. Our comparison tests whether such prompting-based fusion transfers to a broader disjoint-corpus setting with emotion, personality, and ambivalence targets.

B.2 Multi-Task Learning Strategies

Equal weighting assigns the same normalized contribution to all active task losses. In our imple-

Symbol	Meaning	Symbol	Meaning
$\mathcal{T}$	Set of tasks {emo, per, amb}.	emo, per, amb	Emotion, personality, and ambivalence tasks.
$\mathcal{D}_t$	Corpus for task t .	N_t	Number of examples in $\mathcal{D}_t$.
(x_i^t, y_i^t)	Input and target for example i , task t .	$\mathcal{Y}_t$	Target space of task t .
i, t, q	Example, task, and training step/epoch indices.	m, n, k	Modality indices m, n and expert index k .
$\mathcal{M}$	Modality set {T, D, V, A}.	$x_{i,m}^t$	Modality stream m for example i , task t .
Φ_m	Feature extractor for modality m .	$X_{i,m}^t$	Feature matrix in $\mathbb{R}^{L_{i,m}^t \times d_m}$.
$L_{i,m}^t$	Number of tokens, frames, or temporal units.	d_m	Feature dimension of modality m .
$\bar{X}_{i,m}^t$	Mean and STD summary in $\mathbb{R}^{2 \times d_m}$.	$H_{i,m}^t$	Encoded summary in $\mathbb{R}^{2 \times d}$.
d	Shared hidden dimension.	$E_{m \rightarrow n}$	Directed expert from modality m to n .
CrossAttn $_{m \rightarrow n}$	Cross-attention block for ordered pair $m \rightarrow n$.	$U_{i,m \rightarrow n}^t$	Cross-attention output in $\mathbb{R}^{2 \times d}$.
$e_{i,m \rightarrow n}^t$	Output of expert $E_{m \rightarrow n}$.	K	Number of experts; $K = 12$ for four modalities.
$e_{i,k}^t$	Expert output indexed by ordered pair k .	$\mathcal{E}(x_i^t)$	Set of expert outputs for example i , task t .
g_t	Task-specific gate scoring network.	$s_{i,k}^t$	Pre-softmax score for expert k .
$\alpha_{i,k}^t$	Normalized weight of expert k .	z_i^t	Task-specific fused representation.
h_t	Prediction head for task t .	$\hat{y}_i^t$	Prediction for example i , task t .
$\mathcal{L}_t$	Loss function for task t .	$\lambda_t^{(q)}$	Weight of task t at step/epoch q .
$\mathcal{J}(\theta)$	Total MTL loss.	$\mathcal{L}_{\text{total}}^{(q)}$	Total loss including optional FM terms.
$\mathcal{L}_{\text{FM}}^{\text{feat}}$	Feature-level FM loss.	$\mathcal{L}_{\text{FM}}^{\text{pred}}$	Prediction-level FM loss.
$\beta_{\text{feat}}, \beta_{\text{pred}}$	Weights of feature- and prediction-level FM.	γ	FM interpolation coefficient.
$u_{i,t}^0, u_{i,t}^1$	Initial and target endpoints for feature-level FM.	$u_{i,t}^\gamma$	Interpolated feature-level point.
v_ϕ^{feat}	Feature-level FM vector field.	P_t	Projection to latent prediction space.
$r_i^t, r_i^{t,*}$	Projected prediction and projected target.	$r_{i,t}^\gamma$	Interpolated prediction-level point.
v_ψ^t	Task-specific prediction-level FM vector field.	$s_t^{(q)}$	Validation score of task t .
$s_t^*(q)$	Best validation score up to epoch q .	$p_t^{(q)}$	Relative progress of task t .
$\delta_t^{(q)}$	Validation degradation of task t .	$\ell_t^{(q)}$	PAGB task lag.
θ_s	Shared model parameters.	$g_t^{(q)}$	Gradient norm on shared parameters.
$\bar{g}_t^{(q)}$	EMA-smoothed gradient norm.	η_λ, η_g	EMA smoothing coefficients.
κ_p, γ_p	PAGB lag and gradient-correction hyperparameters.	$\tilde{\lambda}_t^{(q)}$	Raw PAGB task weight.
$\lambda_{\min}, \lambda_{\max}$	PAGB clipping bounds.	$\mathcal{R}$	Set of evaluation metrics for ranking.
$\text{rank}_r(a)$	Rank of model a under metric r .	$\text{Rank}_{\text{avg}}(a)$	Average rank across metrics.

Table 8: Summary of the main notation used in the paper.

mentation, if A is the set of active tasks in a batch, each task receives weight $1/|A|$. This is the default reference strategy for evaluating whether adaptive task balancing is necessary.

Learned task weighting treats task weights as trainable parameters. In the implementation, each task has a learnable logit initialized from the logarithm of its fixed loss weight. At each update, the logits of active tasks are normalized with a softmax:

$$\lambda_t = \frac{\exp(\omega_t)}{\sum_{r \in A} \exp(\omega_r)}, \quad t \in A,$$

where A is the set of active tasks in the current batch and ω_t is the learned task logit. This strategy learns static task preferences from the training objective, but unlike PAGB, it does not explicitly use validation progress, degradation, or gradient-norm dynamics.

PCGrad projects task gradients to reduce pairwise conflicts (Yu et al., 2020). It is included as a representative gradient-surgery method for testing whether negative transfer can be reduced by modifying conflicting task updates.

CAGrad searches for a conflict-averse update

direction while preserving convergence toward the average multi-task loss (Liu et al., 2021). It tests whether a conservative shared update improves the trade-off across the three behavioral tasks.

FairGrad formulates MTL strategy as fair resource allocation across tasks (Ban and Ji, 2024). It is relevant because the proposed setting contains tasks with different corpus sizes, losses, and learning dynamics.

NTKMTL addresses task imbalance using convergence behavior motivated by Neural Tangent Kernel analysis (Qin et al., 2025). It is included because it also targets unequal learning speed, but does so through an NTK-inspired balancing mechanism rather than the direct progress and degradation signals used by PAGB.

IMGrad introduces imbalance sensitivity into gradient-based MTL (Zhou et al., 2025b). We include it to test whether imbalance-aware gradient manipulation can match or complement progress-aware task weighting.

PAGB is our progress-aware MTL strategy. It updates task weights using relative task progress, recent validation degradation, and EMA-smoothed gradient norms. Unlike learned weighting, which only learns task logits from the training objective, PAGB explicitly reacts to the current training dynamics of each task.

C Single-Task SOTA Systems

This section summarizes the task-specific systems used in the single-task comparison. Unlike the multimodal fusion baselines in Appendix B, these methods are not adapted to the disjoint-corpus MTL protocol. They are reported as corpus-specific reference points for evaluating whether the unified model remains competitive with systems optimized for one task at a time.

C.1 Emotion Recognition on CMU-MOSEI

MASAI (Markitantov et al., 2026) is a multilingual multimodal multi-task method for emotion and sentiment recognition. It uses audio, video, and text modalities and proposes a triple-fusion strategy to improve intra- and inter-modal interactions. We include it as a strong recent reference for multimodal affective-state recognition on CMU-MOSEI.

MER-CAPF (Chen et al., 2026) is an audio-text emotion recognition model based on cross-attention and multi-granularity pooling. It combines BERT-based textual features with acoustic

representations and uses cross-modal attention to capture semantic-acoustic interactions. We include it as a recent cross-attention-based emotion recognition baseline.

AVTEmo-FNet (Arumugam et al., 2025) combines audio, visual, and textual representations for multimodal emotion recognition. It uses modality-specific encoders and an attention-based fusion mechanism. We include it as a task-specific multimodal emotion recognition system.

C.2 Personality Assessment on FIV2

DLF (Ning et al., 2026) predicts Big Five personality traits using visual and audio features. It applies multi-scale visual decomposition, audio processing, feature alignment, and decision-level fusion. We include it as a recent multimodal personality assessment reference.

MP-PMGRT (Wang et al., 2025b) uses pre-trained multimodal encoders and a graph relational Transformer network for apparent personality prediction. It models intra-modal temporal dynamics and bimodal association features before multimodal attention-based fusion. We include it as a graph-relation-based personality prediction baseline.

MoRE (Zhang et al., 2025) is a structured multi-signal encoder for personality recognition from short media clips. It combines visual, audio, and linguistic signals with adaptive channel selection. We include it as a recent multimodal personality profiling method.

C.3 Ambivalence Recognition on BAH

Divergence-based multimodal fusion (Souza et al., 2026) treats ambivalence as cross-modal incongruence across visual, acoustic, and linguistic modalities. It models this incongruence through pairwise absolute differences between modality embeddings. We include it as a recent BAH reference focused on explicit divergence-based fusion.

ConflictAwareAH (Bekhouché et al., 2026) models ambivalence as a form of cross-modal conflict. It extracts linguistic, acoustic, and visual representations and uses pairwise conflict features to capture disagreement between modalities. This makes it a relevant BAH reference for methods that explicitly emphasize multimodal conflict.

BROTHER (Pereira et al., 2026) uses heterogeneous ensemble regularization for ambivalence recognition. It combines visual, acoustic, linguistic, and statistical speech-related features and uses

Corpus	Task	Labels / targets	Subjects	Women	Age	Train	Dev.	Test	Hours	Mean $\pm$ STD, s	Min-Max, s	Metrics	
CMU-MOSEI	Emotion	Six non-neutral labels: happiness, sadness, anger, fear, disgust, surprise; neutral is derived from the all-zero vector	1,000+	43%	–	16,274	1,861	4,653	45.92	7.28 ± 5.10	0.19–108.95	mean F1-score (mMF1), mean Weighted Accuracy (mWAcc)	Macro
Fliv2	Personality	Big Five traits: OPE, CON, EXT, AGR, NNEU; scores in $[0, 1]$	3,000+	57%	11–60	6,000	2,000	2,000	42.42	15.27 ± 0.48	2.07–15.45	mean F1-score (mMF1), mean Concordance Coefficient (CCC)	Accuracy (mACC), mean Correlation Coefficient (CCC)
BAH	Ambivalence	Binary ambivalence label	300	52%	18–74	776	124	525	10.59	26.78 ± 16.56	3.14–95.66	Macro F1-score (MF1), UAR	

Table 9: Corpora used in the disjoint-corpus multimodal MTL setting. All corpora are English-language. Subjects denotes the number of unique speakers or participants, Women denotes the percentage of female subjects, and Age denotes the reported age range or summary when this information is available.

Category	Parameter	Search values
Input streams	active_modalities	audio, text, scene/visual, behavior-description
Candidate frames	counter_need_frames	30
Feature aggregation	average_features	mean, mean_std_seq, raw
Model	model_name	TACME_v1, TACME_v2, UniGraph2, DPDF-LQ, HME, Comp
Hidden size	hidden_dim (d)	128
Attention heads	num_transformer_heads	8
Attention layers	tf_layers	5
Dropout	dropout	0.15
Output feature size	out_features	128
Active tasks	active_tasks ($\mathcal{T}$)	emotion, personality, ambivalence (internal key: ah)
Task weighting	loss_weighting (λ_t)	equal, learned
MTL strategy	mtl_method	off, ntkmtl, fairgrad, pcgrad, cagrad, imgrad, pagb
FM mode	flow_matching_mode ($\mathcal{L}_{FM}^{feat}, \mathcal{L}_{FM}^{pred}$)	off, features, heads, features+heads
FM steps	flow_matching_steps (S_{FM})	4
FM weight	flow_matching_weight ($\beta_{feat}, \beta_{pred}$)	0.01
PAGB lag sensitivity	pagb_tau (κ_p)	4.0
PAGB gradient correction	pagb_gamma (γ_p)	0.5
PAGB weight bounds	pagb_min_weight (λ_{min})	0.5
	pagb_max_weight (λ_{max})	2.0
PAGB smoothing	pagb_ema (η_λ)	0.5
	pagb_grad_ema (η_g)	0.9
PAGB warmup	pagb_warmup_epochs	1
Batch size	batch_size	32
Epochs	num_epochs	100
Early stopping	max_patience	5
Optimizer	optimizer	adam, adamw, lion, sg, rmsprop
Learning rate	lr	10^{-4}
Weight decay	weight_decay	10^{-5}
Scheduler	scheduler_type	plateau, cosine, onecycle, none

Table 10: Main implementation and training parameters. The table lists the configuration parameters and candidate values used in model comparison and hyperparameter selection. TACME_v1 and TACME_v2 denote the TACME model with equal experts and with task-aware experts, respectively.

a regularized ensemble strategy. We include it as a strong ensemble-based BAH reference.

D Notation Summary

Table 8 summarizes the main mathematical notation used in the problem formulation and method sections. It includes task, corpus, modality, representation, expert, gating, loss, FM, and PAGB-related symbols.

E General Information about Research Corpora

Table 9 highlights the heterogeneity of the experimental setting. The three corpora differ in task type, label representation, duration distribution, corpus size, and evaluation metrics. CMU-MOSEI is the

largest corpus and contains short segments with six emotion-intensity annotations, whereas Fliv2 has nearly fixed-length clips with continuous Big Five trait scores. BAH is much smaller but contains longer and more variable responses for binary ambivalence recognition. These differences make direct loss or metric comparison across tasks inappropriate and motivate the use of task-specific heads, heterogeneous losses, and average-rank model selection.

For CMU-MOSEI, the original annotation is a six-dimensional emotion-intensity vector $a_i^{\text{emo}} \in [0, 3]^6$, whose dimensions correspond to happiness, sadness, anger, fear, disgust, and surprise. The all-zero vector denotes the neutral state. Since our emotion head predicts a normalized distribution

Model	Aggregation	FM setting	MTL strategy	Params, M	Model size, MB	Train time / epoch, min	Total train time, h	Dev. eval. time, s
TACME with equal experts	Mean	Off	Equal	3.63	14.06	4.2	1.49	20.5
TACME with equal experts	Mean + STD	Off	Equal	3.63	14.06	5.7	2.44	24.3
TACME with equal experts	Raw	Off	Equal	3.63	14.06	8.9	3.56	31.6
TACME with equal experts	Raw	PL	Equal	3.74	14.47	9.3	2.19	33.1
TACME with equal experts	Raw	FL	Equal	3.83	14.84	8.9	6.27	28.4
TACME with equal experts	Raw	FL and PL	Equal	3.94	15.25	9.3	6.10	32.5
TACME with task-aware experts	Raw	FL	Equal	3.88	15.04	10.9	3.50	55.1
TACME with task-aware experts	Raw	FL	Learned	3.88	15.04	8.0	2.94	50.3
TACME with task-aware experts	Raw	FL	NTKMTL	3.88	15.04	13.6	4.21	49.9
TACME with task-aware experts	Raw	FL	FairGrad	3.88	15.04	13.9	5.06	50.5
TACME with task-aware experts	Raw	FL	PCGrad	3.88	15.04	14.7	5.87	50.4
TACME with task-aware experts	Raw	FL	CAGrad	3.88	15.04	14.2	4.38	49.4
TACME with task-aware experts	Raw	FL	IMGrad	3.88	15.04	13.9	5.56	49.6
TACME with task-aware experts	Raw	FL	PAGB	3.88	15.04	12.9	3.78	50.1
UniGraph2	Raw	FL	Equal	2.32	8.91	2.0	1.26	22.8
UniGraph2	Raw	FL	PAGB	2.32	8.91	2.8	1.30	15.5
DPDF-LQ	Raw	FL	Equal	2.71	10.44	5.2	2.26	28.3
DPDF-LQ	Raw	FL	PAGB	2.71	10.44	7.4	2.85	31.0
HME	Raw	FL	Equal	5.60	21.61	9.2	3.12	45.2
HME	Raw	FL	PAGB	5.60	21.61	13.2	4.22	44.2
ComP	Raw	FL	Equal	3.34	12.85	5.2	2.28	28.4
ComP	Raw	FL	PAGB	3.34	12.85	7.3	3.39	27.0

Table 11: Computational cost of the compared models and training strategies. Train time per epoch is averaged over completed epochs before early stopping. Total train time includes development and test evaluation during training. Development evaluation time is measured as one full evaluation pass over the development splits of all three corpora, including data loading and metric computation; it is not a pure forward-only inference corpus. FL and PL denote feature-level and prediction-level FM, respectively.

over neutral and the six non-neutral emotions, we convert each annotation into a seven-dimensional soft label $y_i^{\text{emo}} \in [0, 1]^7$. Let $s_i = \sum_{j=1}^6 a_{i,j}^{\text{emo}}$. If $s_i = 0$, we set $y_i^{\text{emo}} = (1, 0, 0, 0, 0, 0, 0)$. Otherwise, the neutral coordinate is set to zero and the six non-neutral emotion intensities are normalized:

$$y_{i,0}^{\text{emo}} = 0, \quad y_{i,j}^{\text{emo}} = \frac{a_{i,j}^{\text{emo}}}{s_i}, \quad j = 1, \dots, 6.$$

This transformation preserves the relative intensities of non-neutral emotions in multi-emotion samples and provides a normalized target for categorical cross-entropy training. During evaluation, mMF1 and mWAcc are computed over the six original non-neutral emotion categories; the neutral coordinate is not included in the reported emotion metrics.

F Implementation Details

We use the same preprocessing, feature extraction, and data splits for all compared models and MTL strategies. We use random seed 42 for all main experiments. To estimate the statistical robustness of the proposed method to random initialization, we additionally repeat the final TACME configuration with four additional random seeds and report the mean and STD over five runs. Detailed multi-seed statistical results are presented in Appendix R.

The input streams are audio, transcript text, VLLM-generated behavioral descriptions, and visual signal information. Audio features are extracted with EmoWav2Vec2 (Wagner

et al., 2023)¹. Transcript and VLLM-generated behavioral-description features are extracted with EmoRoBERTa². Visual features are extracted with Contrastive Language-Image Pre-Training (CLIP) (Radford et al., 2021)³. Spoken transcripts are obtained with Whisper, while Qwen3-VL-4B-Instruct generates the behavioral-description stream as described in Appendix H. EmoRoBERTa is shared across both textual feature streams.

We do not study the relative effectiveness of different pretrained feature extractors in this work. Instead, we rely on established encoders and focus on the proposed disjoint-corpus multi-task fusion, FM, and MTL strategies. The main implementation and training parameters are summarized in Table 10. Not all parameters are swept for every model. Modality fusion architecture- and MTL strategy-specific parameters are kept at the values recommended by the original implementations unless otherwise stated. The same data splits, feature cache, and evaluation scripts are used for all compared methods.

G Computational Cost Analysis

Table 11 reports the computational cost of the compared multimodal fusion models and MTL strategies. All timing measurements were obtained

¹<https://huggingface.co/audeering/wav2vec2-large-robust-12-ft-emotion-msp-dim>

²https://huggingface.co/michellejieli/emotion_text_classifier

³<https://huggingface.co/openai/clip-vit-base-patch32>

from runs on an NVIDIA Tesla V100-SXM2 GPU (32 GB). The implementation uses Python 3.8.20 and the standard PyTorch ecosystem for GPU-based training. We fix random seeds and use a single configuration format for all compared architectures and MTL strategies. The reported training and evaluation times exclude offline feature extraction and VLLM-based behavior-description generation; they measure only training and evaluation over the precomputed feature caches. Importantly, these costs correspond to a single unified model trained to address all three tasks. Training separate task-specific models would require multiple independent training runs. For deployment-oriented comparison, the multi-task TACME core occupies 15.04 MB, whereas three separate single-task TACME cores require approximately $3 \times 14 = 42$ MB. The dominant storage cost is instead the shared pretrained extraction stack: Whisper (1.62 GB), EmoRoBERTa shared across the transcript and behavioral-description streams (332 MB), Qwen3-VL-4B-Instruct (8.89 GB), CLIP (1.82 GB), and EmoWav2Vec2 (1.32 GB), totaling approximately 14.0 GB. Compared with this shared extraction stack, the TACME core is lightweight. Moreover, task-specific SOTA systems often rely on different optimized feature extractors for different tasks and corpora, whereas multi-task TACME uses one shared feature-extraction pipeline and one task model for all three behavioral tasks.

Aggregation mainly affects runtime: Mean and Mean + STD reduce TACME’s per-epoch time from 8.9 min to 4.2 and 5.7 min without changing the parameter count. Adding prediction-level FM slightly increases the number of parameters from 3.63M to 3.74M, while feature-level and combined feature- and prediction-level FM increase it to 3.83M and 3.94M, respectively. Across multi-modal fusion models, UniGraph2 is the lightest and fastest model, with 2.32M parameters, an 8.91 MB checkpoint, and 2.0 min per epoch. DPDF-LQ and ComP are also compact, while HME is the largest baseline with 5.60M parameters and a 21.61 MB checkpoint.

Task-aware TACME remains moderately sized at 3.88M parameters and 15.04 MB, but is slower than DPDF-LQ and ComP because directed modality-pair experts require additional cross-modal attention. Among MTL strategies, Learned weighting is the cheapest adaptive strategy for task-aware TACME, reducing per-epoch time to 8.0 min in this run. Gradient-based methods are more expen-

sive, requiring 13.6–14.7 min per epoch. PAGB requires 12.9 min per epoch, making it cheaper than the gradient-projection and gradient-conflict baselines while still adding overhead relative to Equal weighting. Across the Equal/PAGB model pairs for UniGraph2, DPDF-LQ, HME, and ComP, PAGB increases per-epoch training time by about 41.8% and total training time by about 27.9% on average. With $|\mathcal{M}| = 4$, TACME uses 12 directed experts; in general, the bank grows as $|\mathcal{M}|(|\mathcal{M}| - 1)$. This remains practical in the present four-modality setting, but larger modality sets may require sparse expert selection, shared expert parameterization, or expert pruning.

H VLLM Behavior Description Prompt

We use a neutral prompt to obtain VLLM-generated behavior descriptions. The descriptions are generated with Qwen3-VL-4B-Instruct⁴. For each video, frames are sampled at 8 frames per second with an upper limit of 16 frames. This preserves short-term behavioral changes while keeping the VLLM input computationally bounded and comparable across corpora with different clip durations. The prompt focuses on observable cues and explicitly avoids inferring target labels such as emotion, personality, or ambivalence. We do not use generated descriptions as labels; they are treated only as an auxiliary textual modality. Prompt wording may affect this modality, but a systematic prompt study is outside the scope of this work. The generated text is used as the D modality. The exact prompt is shown in Figure 3.

You are analyzing a short video clip of a person. Describe only observable behavior: facial movements, gaze direction, head motion, body posture, hand gestures, speaking-related movements, and relevant scene context. Note how these visible cues change over time when possible. Do not infer emotions, personality traits, hesitation, confidence, intentions, mental states, or any other hidden attributes. Write one coherent paragraph with no line breaks, no bullets, no special formatting, no longer than 100 tokens, and always end with a period.

Figure 3: Prompt used for behavior-description extraction.

I VLLM Description Audit

We audit the generated behavioral descriptions for three failure modes: direct target-label leakage,

⁴<https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct>

mental-state inference, and demographic or stereotypical language. Table 12 reports four representative examples from this analysis.

The descriptions mainly encode observable visual evidence, including gaze, mouth movement, gestures, posture, object interaction, and scene context. No direct emotion, apparent personality, or ambivalence labels are exposed in the inspected examples. The audit identifies only one mild behavioral interpretation and two gendered person references, with no stereotypical claims observed. Similar patterns are found in the other inspected descriptions, which predominantly remain grounded in observable behavior and scene content. Accordingly, we use the description stream as an auxiliary natural-language modality that summarizes observable human behavior and scene context and complements the other modalities.

J Demographic Subgroup Analysis

Table 13 reports subgroup performance for the final multi-task TACME model across gender, age, and ethnicity. Demographic metadata are available for the FIV2 and BAH test subsets.

Gender differences are small on FIV2, whereas on BAH the male subgroup achieves higher MF1 and UAR than the female subgroup. Across age groups, FIV2 shows similar mACC but larger variation in CCC, while BAH performance is lowest for the ≤ 24 group. Ethnicity shows the largest variation on FIV2 CCC, ranging from 59.74 to 70.70, while the differences in mACC are much smaller. On BAH, ethnicity-related differences are more moderate across MF1 and UAR. Although subgroup sizes are highly unequal, higher performance is not consistently associated with larger groups: for example, the small Asian subgroup on FIV2 achieves the highest mACC, while the largest age group on BAH (25–44) is not the best-performing group. Thus, the results do not show a consistent tendency for TACME to favor majority subgroups. Because some subgroups contain relatively few samples, these results should be interpreted as descriptive rather than as a definitive fairness assessment.

K Additional Aggregation Results

Table 14 compares feature aggregation strategies across all fusion models. Mean + STD aggregation gives the best average rank for DPDF-LQ, ComP, and both TACME variants, showing that preserv-

ing both first- and second-order feature statistics is generally more effective than using mean-only summaries. The effect is especially clear for TACME: with equal experts, Mean + STD improves the average rank from 7.50 to 4.67, and with task-aware experts from 4.50 to 2.83. This supports our choice of two-token statistical summaries, which reduce variable-length modality streams while retaining information about feature dispersion. HME is the only exception, where mean aggregation performs best, suggesting that its hyper-modality enhancement module is less dependent on explicit variance information. In contrast, using unaggregated inputs is not consistently beneficial and often worsens the rank, especially for DPDF-LQ, ComP, and TACME. Although Mean + STD discards fine-grained temporal ordering, its consistently strong performance across several fusion models indicates that compact first- and second-order statistical summaries provide an effective balance between information retention and sequence reduction in the evaluated setting. Overall, Mean + STD provides the most stable trade-off across tasks and is therefore used in the main TACME configuration.

L Additional Flow Matching Results

Table 15 evaluates different FM configurations for each multimodal fusion model. Overall, adding FM improves the average rank for most models, but the best level of regularization is model-dependent. UniGraph2, DPDF-LQ, ComP, and task-aware TACME obtain their best average ranks when feature- and prediction-level FM are combined, indicating that regularizing both fused representations and task outputs can improve the cross-task trade-off. For HME and TACME with equal experts, feature-level FM performs best, suggesting that these architectures benefit more from stabilizing the shared representation than from additional prediction-space regularization. The strongest result is achieved by task-aware TACME with combined feature- and prediction-level FM, which reduces the average rank from 7.00 to 3.50. This confirms that FM is most effective when paired with task-specific expert weighting, because the model can regularize shared cross-modal representations while still selecting different modality relations for different behavioral tasks.

Example	Full VLLM-generated behavioral description	Target-label leakage	Mental-state inference	Demographic / stereotypical language
1	The person speaks directly to the camera, occasionally gesturing with both hands, sometimes raising them or moving them side to side. Their head tilts slightly and turns as they speak, maintaining eye contact with the camera. They adjust their hair with one hand at one point. The background includes a black office chair, a TV, and framed pictures on the wall. The person remains seated throughout, with their upper body slightly forward, indicating engagement with the viewer. Their mouth moves consistently as they speak, and.	No	Minimal: "indicating engagement" is a mild behavioral interpretation	No
2	The man speaks directly to the camera, his mouth moving and eyes focused forward, occasionally blinking. His head remains mostly still with slight nods. He wears a gray t-shirt with a Mickey Mouse graphic and "Walt Disney World" text. His hands are mostly stationary, but he makes minor gestures with his right hand. Behind him, three aquariums with green plants and water are visible, and a framed picture hangs on the wall above them. The lighting is consistent, and the background remains unchanged throughout.	No	No	Minimal: gendered person reference
3	The man adjusts his earpiece with his right hand, then lowers it, maintaining eye contact with the camera while speaking. His head slightly tilts and turns as he speaks, and his mouth moves rhythmically. He remains seated, with his shoulders relaxed, and his posture steady throughout the clip. The background features a brick wall and a concrete wall with partial lettering. He continues speaking without significant changes in expression or movement.	No	No	Minimal: gendered person reference
4	The person reads from a book, looking down at the pages, then lifts their gaze to face forward while closing the book and placing it down. Their head tilts slightly as they speak, and their hands move from holding the book to resting on their lap. The background remains static with a couch and a white wall visible. The person maintains a steady posture throughout.	No	No	No

Table 12: Audit of the inspected VLLM-generated behavioral descriptions. The descriptions are treated as an auxiliary natural-language modality rather than labels.

Attribute	Group	FIV2			BAH		
		# videos	mACC $\uparrow$	CCC $\uparrow$	# videos	MF1 $\uparrow$	UAR $\uparrow$
Gender	Male	888	91.70	68.82	266	73.33	74.15
Gender	Female	1112	91.73	69.90	252	69.95	70.31
Age	≤ 24	267	92.10	68.01	91	66.51	66.61
Age	25–44	1527	91.71	69.23	343	72.14	73.00
Age	≥ 45	206	91.24	71.94	91	74.53	74.75
Ethnicity	Asian	48	92.95	60.59	140	68.58	70.78
Ethnicity	Caucasian / White	1728	91.73	70.70	231	72.67	72.67
Ethnicity	Afro-American / Black	224	91.36	59.74	63	74.95	74.58
Ethnicity	Mixed	–	–	–	70	70.56	74.28
Ethnicity	Other	–	–	–	21	74.07	73.56

Table 13: Demographic subgroup performance of the final multi-task TACME model on the FIV2 and BAH test subsets.

Model	Aggregation	CMU-MOSEI		FIV2		BAH		Avg. Rank $\downarrow$
		mMF1 $\uparrow$	mWAcc $\uparrow$	mACC $\uparrow$	CCC $\uparrow$	MF1 $\uparrow$	UAR $\uparrow$	
UniGraph2	Mean	59.42	60.29	90.62	54.42	67.38	68.89	13.17
	Mean + STD	60.28	60.90	90.56	54.38	67.35	68.37	13.50
	Raw	60.17	61.13	90.48	52.45	68.50	68.31	13.00
DPDF-LQ	Mean	61.34	62.29	90.90	57.73	69.09	69.59	8.17
	Mean + STD	62.05	62.76	90.99	57.95	68.00	69.18	6.17
	Raw	62.18	62.69	90.91	59.36	65.18	66.05	10.00
HME	Mean	61.92	62.59	91.51	64.63	66.62	66.95	7.83
	Mean + STD	61.06	62.13	91.36	61.49	67.66	68.07	10.67
	Raw	61.27	62.35	91.37	63.94	67.07	66.96	10.50
ComP	Mean	59.24	61.25	90.52	50.57	67.29	69.15	14.33
	Mean + STD	59.17	61.56	90.52	50.82	69.32	69.47	11.50
	Raw	59.39	61.90	90.36	50.69	66.18	66.64	16.33
TACME with equal experts	Mean	61.56	62.16	91.42	66.48	67.81	68.18	7.50
	Mean + STD	61.76	62.50	91.28	64.92	69.06	70.02	4.67
	Raw	61.47	62.15	91.38	66.45	67.50	67.63	9.17
TACME with task-aware experts	Mean	61.54	62.44	91.47	67.66	68.29	69.72	4.50
	Mean + STD	61.88	62.45	91.58	67.73	69.05	70.01	2.83
	Raw	61.39	62.30	91.47	67.13	67.69	68.71	6.83

Table 14: Comparison of feature aggregation strategies for multimodal fusion models. All variants use equal weights for task-specific losses. **Best** and **second-best** results are highlighted.

Model	Aggregation	FM setting	CMU-MOSEI		Fiv2		BAH		Avg. Rank ↓
			mMF1 ↑	mWAcc ↑	mACC ↑	CCC ↑	MF1 ↑	UAR ↑	
UniGraph2	Mean	W/o FM	59.42	60.29	90.62	54.42	67.38	68.89	20.83
		Feature-level	60.07	61.47	91.28	58.78	68.36	68.65	16.67
		Prediction-level	60.33	60.96	91.38	58.89	67.66	69.46	15.83
		Feature- and prediction-level	60.41	61.15	91.23	61.53	68.90	68.97	15.17
DPDF-LQ	Mean + STD	W/o FM	62.05	62.76	90.99	57.95	68.00	69.18	13.00
		Feature-level	62.18	62.69	91.23	61.10	68.08	69.05	11.67
		Prediction-level	62.03	62.56	90.98	58.76	68.36	69.14	12.83
		Feature- and prediction-level	62.45	62.89	90.90	58.15	68.84	69.71	9.33
HME	Mean	W/o FM	61.92	62.59	91.51	64.63	66.62	66.95	14.00
		Feature-level	62.10	62.30	91.47	65.08	67.64	69.38	11.00
		Prediction-level	62.07	62.26	91.53	64.55	66.87	67.72	14.17
		Feature- and prediction-level	62.09	62.30	91.53	65.14	67.23	67.59	12.83
Comp	Mean + STD	W/o FM	59.17	61.56	90.52	50.82	69.32	69.47	16.67
		Feature-level	59.90	61.71	90.48	51.21	68.21	69.07	18.67
		Prediction-level	59.39	61.90	90.48	51.47	67.82	69.54	18.17
		Feature- and prediction-level	60.56	61.81	90.48	51.72	68.90	69.96	14.83
TACME with equal experts	Mean + STD	W/o FM	61.76	62.50	91.28	64.92	69.06	70.02	8.67
		Feature-level	62.31	62.77	91.61	69.17	67.90	69.26	6.17
		Prediction-level	61.88	62.56	91.43	66.59	68.18	68.91	11.00
		Feature- and prediction-level	61.75	62.46	91.32	64.20	67.14	67.88	15.83
TACME with task-aware experts	Mean + STD	W/o FM	61.88	62.45	91.58	67.73	69.05	70.01	7.00
		Feature-level	62.48	63.06	91.37	66.14	69.37	69.85	4.33
		Prediction-level	62.03	62.50	91.56	68.34	69.06	70.02	5.33
		Feature- and prediction-level	62.25	62.80	91.60	68.51	68.74	70.37	3.50

Table 15: Comparison of FM configurations for multimodal fusion models. Feature-level FM regularizes fused representations, while prediction-level FM regularizes task-specific outputs. All variants use equal weights for task-specific losses. **Best** and **second-best** results are highlighted.

Variant	CMU-MOSEI		Fiv2		BAH		Avg. Rank ↓
	mMF1 ↑	mWAcc ↑	mACC ↑	CCC ↑	MF1 ↑	UAR ↑	
Full PAGB	62.68	63.29	91.71	69.89	71.15	72.07	2.08
w/o progress lag	62.54	62.87	91.73	70.11	69.50	71.29	2.50
w/o degradation term	62.43	62.89	91.77	70.46	68.62	70.00	3.33
w/o gradient correction	62.50	62.86	91.52	67.97	69.00	70.46	4.58
w/o weight smoothing	62.48	62.74	91.72	69.88	69.83	71.57	3.83
w/o clipping	62.50	62.82	91.71	69.98	68.29	69.72	4.67

Table 16: Ablation of the proposed PAGB strategy. All variants use task-aware TACME with Mean + STD aggregation and the feature- and prediction-level FM setting. Each row disables one component of PAGB. **Smallest** and **largest** degradations are highlighted.

M Task-Dependent Input Attention

Figure 4 complements the average expert-weight analysis in Figure 2 with sample-level visualizations. While Figure 2 shows which directed modality-pair experts are preferred on average, Figure 4 shows how the final model distributes attention over the input statistical tokens for individual test examples. Each row corresponds to one sample, and the heatmaps report task-specific attention over the Mean and STD tokens of the transcript T, VLLM-generated behavioral description D, visual signal V, and acoustic signal A.

The examples show that the same multimodal input is used differently by different task heads. For the happiness example, the heatmaps assign strong responses to transcript and acoustic statistics for emotion and personality outputs, while the ambivalence head focuses on a different subset of text and behavioral-description tokens. In the fear and

ambivalence example, the ambivalence prediction relies more strongly on acoustic and description-related statistics, which is consistent with the high predicted ambivalence probability. The anger and sadness examples show further task-dependent redistribution of attention: emotion heads emphasize different combinations of visual, textual, and acoustic statistics, whereas personality heads use broader cross-modal patterns. These qualitative patterns support the main assumption behind TACME: task-specific prediction cannot be reduced to a single fixed fusion pattern, because emotion, personality, and ambivalence attend to different modality statistics even for the same input sample.

N Loss Convergence

Figure 5 compares development-loss dynamics for different MTL strategies. Fiv2 shows the most stable convergence: most strategies decrease rapidly



Figure 4: Sample-level task-dependent input-attention visualizations for four test examples. In each row, the left panel shows the four input streams: transcript T, VLLM-generated behavioral description D, visual signal V, and acoustic signal A. The right panel shows task-specific attention maps over the Mean and STD statistical tokens of each modality.

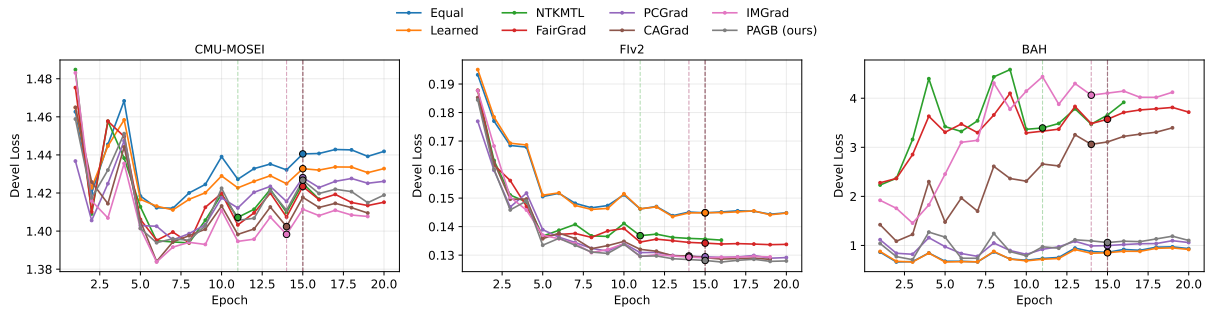


Figure 5: Development-loss convergence under different MTL strategies. The curves compare Equal weighting, Learned weighting, NTKMTL, FairGrad, PCGrad, CAGrad, IMGrad, and PAGB using the same task-aware TACME configuration.

in the first epochs and then remain in a narrow loss range. CMU-MOSEI is less stable, with mild oscillations after the initial decrease. In contrast,

BAH acts as the most challenging task: several recent gradient-based strategies show high or unstable BAH losses, indicating that improvements

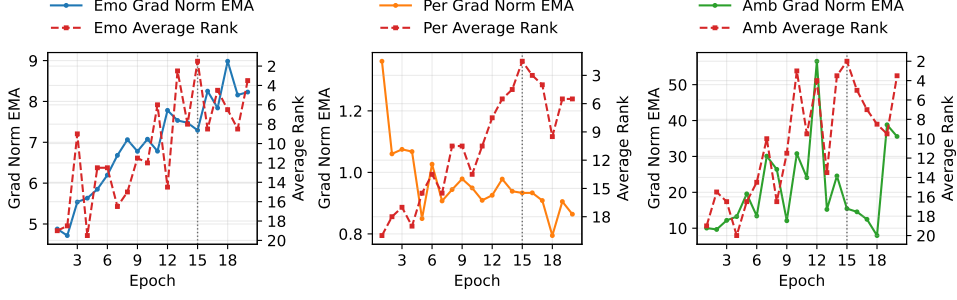


Figure 6: EMA-smoothed task gradient norms $\bar{g}_t^{(q)}$ and task-specific average validation ranks during PAGB training. The gray vertical line marks the selected checkpoint epoch q^* .

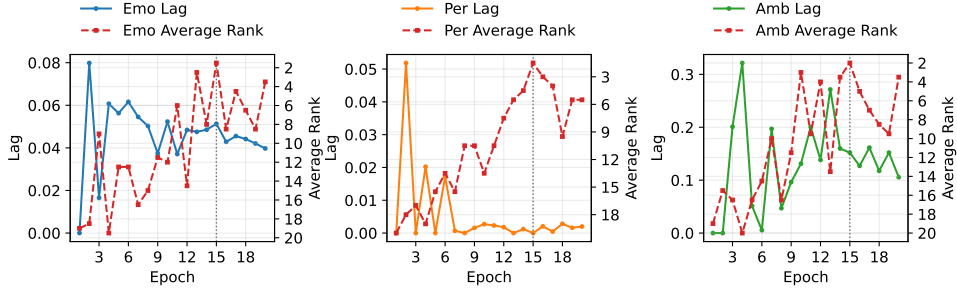


Figure 7: Task lag values $\ell_t^{(q)}$ and task-specific average validation ranks during PAGB training. The gray vertical line marks the selected checkpoint epoch q^* .

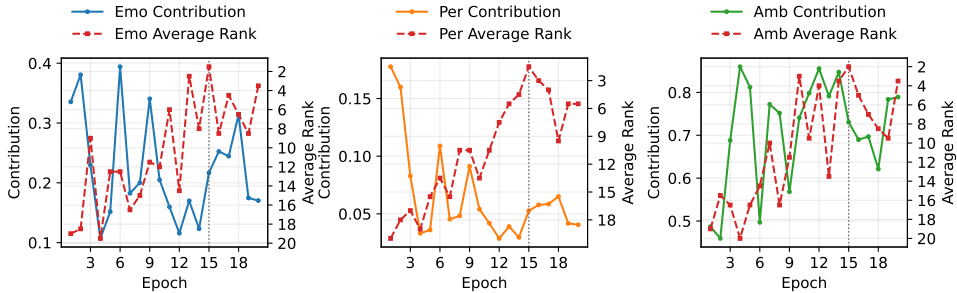


Figure 8: Effective task contributions, corresponding to the normalized task weights $\lambda_t^{(q)}$, and task-specific average validation ranks during PAGB training. The gray vertical line marks the selected checkpoint epoch q^* .

on the larger emotion and personality corpora can come at the cost of ambivalence recognition. More conservative strategies, such as Equal, Learned, PCGrad, and CAGrad, avoid the strongest BAH degradation, but they do not provide the best aggregate trade-off. PAGB keeps BAH loss in the lower range while maintaining competitive convergence on CMU-MOSEI and Fiv2. This supports the main result in Table 2: PAGB does not over-optimize one task at the expense of another, but instead balances the three heterogeneous corpora more effectively. These dynamics further justify validation average rank as the model-selection criterion, because raw losses have different scales and do not directly re-

flect the cross-task trade-off.

O PAGB Component Ablation

Table 16 ablates the components of PAGB. All variants use task-aware TACME, Mean + STD aggregation, and the feature- and prediction-level FM setting. The full configuration gives the best average rank among the tested variants. Removing gradient correction or clipping causes the largest degradation, which shows that controlling gradient-scale imbalance and bounding task weights are important for stable disjoint-corpus optimization. Removing the degradation term or weight smoothing also worsens the average rank, although the effect

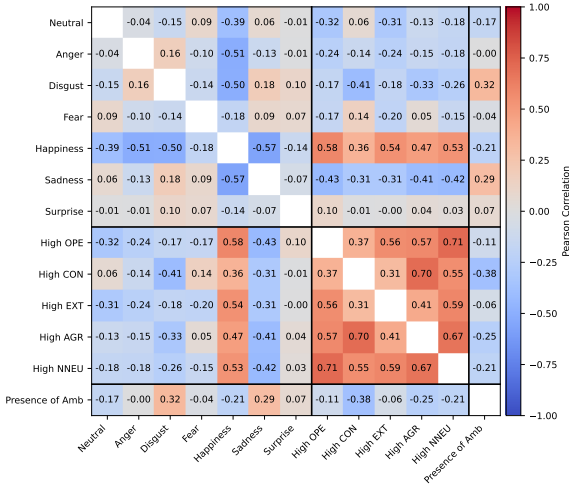


Figure 9: Pearson correlation matrix over pooled predictions from the trained TACME model. The matrix includes emotion probabilities, high-trait indicators obtained by thresholding personality scores at 0.5, and ambivalence-presence probabilities.

is smaller. Removing only the progress-lag term leads to a moderate drop, suggesting that this signal refines the weighting dynamics but is less critical than gradient correction and clipping. The variant that removes progress lag, degradation, and smoothing together performs worse than full PAGB, which indicates that these validation-based and smoothing components remain useful when combined with bounded gradient-aware weighting.

P Training Dynamics

This section analyzes the training dynamics of PAGB. Figure 6 shows that the tasks have different gradient-norm scales. Ambivalence has the largest and most unstable gradient-norm EMA, emotion has a moderate but increasing gradient norm, and personality has the smallest gradient norm after the first epochs. Around the selected checkpoint, the personality and ambivalence average ranks are close to their best values, while emotion remains more variable. This pattern motivates the gradient-norm correction term in PAGB: equal task weights would ignore the large difference in shared-gradient scales.

Figure 7 reports the task lag signal. Personality has near-zero lag after the early epochs, which indicates stable progress under the shared representation. Ambivalence has the largest and most fluctuating lag, reflecting less stable validation behavior. Emotion lies between these two tasks. Near the selected checkpoint, the lag signal identifies

which tasks need additional optimization effort, while the average rank curves show that this effort should be balanced against the aggregate validation trade-off rather than a single task metric.

Figure 8 shows the effective task contributions after lag-based weighting, gradient-norm correction, clipping, and smoothing. Ambivalence receives the largest contribution during most epochs because it has high lag and unstable optimization dynamics. Personality receives the smallest contribution because its lag is low and its average rank is more stable. Emotion receives an intermediate contribution, with fluctuations that follow changes in average rank and gradient-norm EMA. Around the selected checkpoint, PAGB therefore allocates more optimization weight to the task that is hardest to stabilize while avoiding overemphasis on the task that already progresses smoothly.

Q Task Correlation Analysis

Figure 9 shows Pearson correlations between predicted task outputs produced by the final TACME model. For this analysis, we run the trained model on all test samples and obtain predictions for all task heads, regardless of the corpus from which a sample originates. The correlation matrix shows that the tasks are related but not redundant. Personality traits form the strongest positive block, with high OPE, EXT, AGR, and NNEU, showing correlations up to 0.71. Happiness is positively correlated with several high personality-trait indicators, whereas sadness is negatively correlated with them. Ambivalence shows a different profile: it is positively correlated with disgust and sadness, and negatively correlated with CON, AGR, NNEU, and happiness. These correlations are moderate rather than extreme, suggesting that ambivalence is not merely a negative-emotion proxy. The analysis therefore supports the disjoint-corpus multi-task setup: the tasks share behavioral cues but require separate prediction heads and task-specific cross-modal weighting.

R Multi-Seed Statistical Results

Table 17 reports multi-seed results for single-task, pairwise dual-task, and full multi-task TACME. The main experiments in the paper use seed 42, while the additional runs estimate robustness to random initialization.

Single-task TACME achieves the highest task-specific scores on all corpora, as each model is opti-

Method	Run	CMU-MOSEI		Fiv2		BAH	
		mMF1 $\uparrow$	mWAcc $\uparrow$	mACC $\uparrow$	CCC $\uparrow$	MF1 $\uparrow$	UAR $\uparrow$
Single-task TACME	1	67.65	68.20	92.78	78.23	77.93	77.78
Single-task TACME	2	68.02	67.97	92.42	77.66	78.91	79.70
Single-task TACME	3	68.37	68.35	92.80	78.89	77.99	78.41
Single-task TACME	4	67.55	68.71	92.47	76.28	77.19	78.12
Single-task TACME	5	67.51	67.60	92.48	76.29	78.25	79.14
	Mean $\pm$ STD	67.82 $\pm$ 0.37	68.17 $\pm$ 0.42	92.59 $\pm$ 0.18	77.47 $\pm$ 1.17	78.05 $\pm$ 0.62	78.63 $\pm$ 0.78
Dual-task TACME	1	63.35	64.32	91.92	72.67	–	–
Dual-task TACME	2	63.04	64.26	91.90	72.57	–	–
Dual-task TACME	3	63.24	63.90	91.89	72.85	–	–
Dual-task TACME	4	62.90	63.78	92.01	72.96	–	–
Dual-task TACME	5	63.11	63.60	92.16	73.34	–	–
	Mean $\pm$ STD	63.13 $\pm$ 0.17	63.97 $\pm$ 0.31	91.98 $\pm$ 0.11	72.88 $\pm$ 0.30	–	–
Dual-task TACME	1	62.72	63.17	–	–	73.52	73.94
Dual-task TACME	2	62.94	63.38	–	–	74.28	74.61
Dual-task TACME	3	63.43	63.95	–	–	72.94	74.64
Dual-task TACME	4	63.14	63.54	–	–	74.73	75.05
Dual-task TACME	5	63.21	63.76	–	–	73.25	73.66
	Mean $\pm$ STD	63.09 $\pm$ 0.27	63.56 $\pm$ 0.31	–	–	73.74 $\pm$ 0.74	74.38 $\pm$ 0.57
Dual-task TACME	1	–	–	91.72	73.17	73.24	73.76
Dual-task TACME	2	–	–	91.91	71.98	73.78	74.23
Dual-task TACME	3	–	–	92.02	72.53	74.13	74.67
Dual-task TACME	4	–	–	91.83	71.26	74.46	75.09
Dual-task TACME	5	–	–	91.89	73.01	74.51	75.11
	Mean $\pm$ STD	–	–	91.87 $\pm$ 0.11	72.39 $\pm$ 0.78	74.02 $\pm$ 0.53	74.57 $\pm$ 0.58
Multi-task TACME	1	62.68	63.29	91.71	69.89	71.15	72.07
Multi-task TACME	2	62.87	63.46	91.81	70.85	70.49	72.13
Multi-task TACME	3	62.59	63.58	91.72	69.13	69.12	70.83
Multi-task TACME	4	62.32	62.94	91.83	70.99	71.07	71.88
Multi-task TACME	5	62.25	63.33	91.75	69.83	69.60	71.66
	Mean $\pm$ STD	62.54 $\pm$ 0.26	63.32 $\pm$ 0.24	91.76 $\pm$ 0.05	70.14 $\pm$ 0.78	70.29 $\pm$ 0.90	71.71 $\pm$ 0.53

Table 17: Multi-seed results over five runs with different random seeds for single-task, pairwise dual-task, and full multi-task TACME. The summary row reports mean $\pm$ STD.

mized for one corpus, one label space, and one evaluation objective. Pairwise dual-task training gives an intermediate result between single-task specialization and full three-task learning. These results indicate a clear trade-off between task-specific specialization and cross-task generality. Reducing the number of jointly optimized tasks alleviates negative transfer and improves the corresponding task metrics. The full multi-task model faces the hardest setting, since it must balance three independent corpora with different label spaces, losses, modality relevance patterns, and evaluation metrics.

The results are stable across random seeds. For single-task TACME, the STDs are low on CMU-MOSEI mMF1/mWAcc (0.37/0.42), Fiv2 mACC (0.18), and BAH MF1/UAR (0.62/0.78), while Fiv2 CCC varies more strongly (1.17). The dual-task variants show low-to-moderate variation, with STDs below 0.31 for CMU-MOSEI, below 0.78 for Fiv2, and below 0.74 for BAH. For full multi-task TACME, the STDs remain low on CMU-MOSEI mMF1/mWAcc (0.26/0.24) and

Fiv2 mACC (0.05), while higher variability appears for Fiv2 CCC (0.78) and BAH MF1/UAR (0.90/0.53).

Overall, single-task TACME is best for task-specific specialization, pairwise dual-task training reduces part of the negative transfer observed in the full setting, and full multi-task TACME provides one unified model for all three heterogeneous behavioral objectives. The full model is therefore evaluated primarily by its cross-task trade-off rather than by maximum performance on each individual corpus.